TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY



Final Year Project Report on

"कानुन" - APPLICATION FOR PERFORMING RAG OPERATION ON CONSTITUTION OF NEPAL

For the partial fulfillment of the requirements of the degree of
Bachelor of Science in Computer Science and Information Technology (B.Sc.CSIT)
awarded by Tribhuvan University

Submitted By:

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SUPERVISOR RECOMMENDATION

This is to certify that the final year project entitled **"कानुन" - APPLICATION FOR PERFORMING RAG OPERATION ON CONSTITUTION OF NEPAL** is an academic work completed by **Abhinab Shrestha (T.U. Exam Roll No. 25818/77)** and **Manish Kumar Rajbanshi (T.U. Exam Roll No. 27251/077)** submitted in the partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Information Technology awarded by Institute of Science and Technology, Tribhuvan University under my guidance and supervision.

The information presented by them in the project report has not been submitted earlier to the best of my knowledge.

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The undersigned certify that they have read and recommended to the Department of Computer Science, St. Xavier's College for acceptance, a Final Year Project Report entitled **"कानुन" - APPLICATION FOR PERFORMING RAG OPERATION ON CONSTITUTION OF NEPAL** submitted by **Abhinab Shrestha (T.U. Exam Roll No. 25818/77)** and **Manish Kumar Rajbanshi (T.U. Exam Roll No. 27251/077)** for the partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science and Information Technology awarded by Tribhuvan University.

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ABSTRACT

The Constitution is the foundation of a nation's legal system, justice, equality, and the protection of citizens' rights. Despite its significance, the complexity of constitutional language often creates a knowledge gap, leaving citizens, legal professionals, and government officials struggling to access and comprehend key provisions. "कानुन" (Kanun) is an LLM based Chat Application built on Open AI's "gpt-4o-mini-2024-07-18" model. This innovative platform is designed to understand Nepal's Constitution which provides accurate, accessible, and contextually relevant information. Through advanced RAG techniques, "कानुन" bridges the gap between legal language and its users, empowering them with knowledge about their constitutional rights and responsibilities.

The application employs a user-friendly interface and robust query-processing capabilities to deliver precise answers to user inquiries. By leveraging research in AI and legal informatics, the system is tailored to address the challenges of interpreting legal texts while maintaining high levels of accuracy and reliability. This report details the problem statement, objectives, and methodology behind the development of "कानुन" It also explores the potential of LangChain to transform legal services by making constitutional knowledge more accessible, promoting legal literacy and encouraging adherence to constitutional principles.

Keywords*:* *LangChain, Constitution, Legal Literacy, Natural Language Processing (NLP), AI Chat Application, Nepal*

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LIST OF ABBREVIATIONS

API	Application Programming Interface
LLM	Large Language Model
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
DBMS	Database Management System
IG	Information Gain
JSON	JavaScript Object Notation
SQLite3	Standard Query Language

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The Constitution is a basic principle or law of a country that citizens of the country guide how everything works and ensure justice for everyone. The Constitution is a fundamental requirement for citizens, legal professionals, and government officials hence it helps protect their human rights and responsibilities. Although complication in legal language can create a knowledge gap and make it difficult for the citizens of the country to access and interpret constitutional information[1].

"कानुन" (KANUN) is a Retrieval-Augmented Generation (RAG)-based system designed to simplify access to constitutional information with precision and reliability. Large Language Models (LLMs) had an advanced retrieval framework through its capabilities helps in max utilization. "कानुन" ensures that the information provided is not only relevant but also rooted in authentic sources, such as the Constitution of Nepal. The Traditional AI models that depend mainly on pre-existing data where "कानुन" diligently searches for and retrieves relevant portions of the constitution before creating responses. This method improves accuracy, keeps legal classification aligned with official documents and gives users clear and accurate information about constitutional provisions.

Using retrieval methods and advanced language models, "कानुन" gives a well-structured and clear solution for exploring constitutional law. Its ability to provide direct and accurate references makes it a valuable tool for legal experts, students, and anyone pursuing clarity on Constitutional law of Nepal.

1.2 Problem Statement

The Constitution act is the backbone of the legal system serving the rights of the citizens, ensuring justice and responsibilities. Constitution plays a vital role in governing the country as the rules and regulations are based on this framework.

However, accessing and understanding constitution acts is quite challenging due to its following reasons:

1. complexity structure of legal documents and which leads to a knowledge gap where citizens are not fully aware of their constitutional acts rights and responsibilities.

2. Absence of an accessible and interactive tool to address constitution related queries where, citizens will be unaware of their constitutional rights and legal actions.

1.3 Objective

The major objective of the "कानुन" (KANUN) is to create an interactive framework for making an interactive application based on RAG which is able to generate information of Constitution. After the successful completion of the project, this project aims to achieve the following objectives:

1. To parse the constitutional PDF data into JSON format consisting of parts, articles, clauses, sub-clauses, explanation by matching regex pattern.
2. To provide a secured and user friendly interface for prompting queries regarding the constitution of Nepal.
3. To provide an information interface for understanding the functioning of LLM operation related to RAG operation happening in real time.

1.4 Scope and Limitations

1.4.1 Scope

The scope of the system includes:

- The project focuses on developing Kanun applications, especially for users to easily search and retrieve constitutional laws.
- The system consists of two features: file injection and image generation, which converts pdf files into JSON and text inputs into image generation.
- The primary goal is to assist citizens and government officials to understand their rights and responsibilities without the complexity of legal documents.
- The user-friendly interfaces were designed for citizens, students and government officials to be easily accessible without any struggle.

1.4.2 Limitation

The limitation of the system include:

- The system fails to understand questions or provides incomplete information due to complex legal language.
- The application is not suitable for professional legal advice but can be used as a guide.

- There are no accessible features for the disabled users in the project such as screen readers or voice commands.
- The system does not include any data privacy or security measures for sensitive legal information.

1.5 Development Methodology

This application uses an Incremental development model, which means the project is developed step by step. First and foremost a simple version is created and shared with the citizen for feedback. Through their input, the application is improved and updated step by step until it becomes fully functional and meets the needs of the citizens.[2]

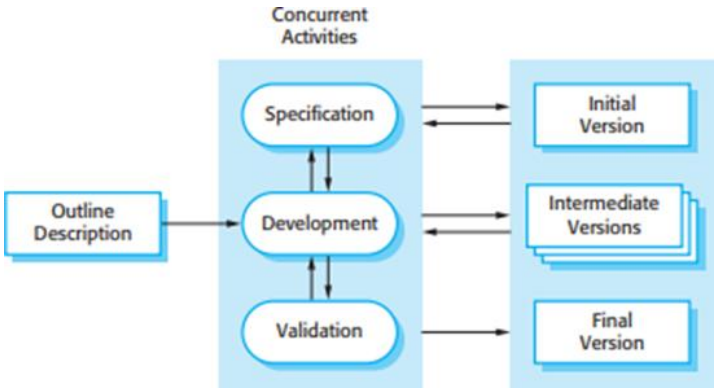


Figure 1: Incremental Development

1 Requirement Analysis:

This is the first phase of the incremental model, the developer identified the requirements. And the system functional requirements are understood. To develop the software under the incremental model, this phase performs a crucial role

2 Design and Development:

In this phase of the Incremental model of SDLC, the design of the system functionality and the development method are finished with success. When software develops new practicality, the incremental model uses style and development phase.

3 Testing:

In the incremental model of the SDLC, the testing phase checks the performance of each existing function as well as additional functionality. In the testing phase, the various methods are used to test the behavior of each task.

4 Implementation:

Implementation phase of SDLC, where the actual coding of the system takes place. It involves scripting the final code based on design from the initial phase and testing the functionality in the testing phase. After this phase, the number of the product working is improved and upgraded up to the final system product

1.6 Report Organization

The report contains details about "कानुन"-application for performing RAG operation on Constitution of Nepal in a structured manner i.e. they are divided into chapters and each chapter presents different aspects involved during the research and development of the project. This project contains 6 chapters.

1. **Chapter 1: Introduction**
   This chapter if for a short introduction about the project, the objectives of the project as well as a problem statement for the project. The scope and limitations for the project are also written in the chapter.
2. **Chapter 2: Background Study and Literature Review**
   This chapter examines the project's background and covers the various research papers related to the project and similar existing systems that already existed.
3. **Chapter 3: System Analysis**
   This chapter introduces a detailed analysis of the system requirements: functional and non-functional requirements along with a feasibility analysis of the system.
4. **Chapter 4: System Design**
   This chapter consists of details about the system design using use case diagrams, class diagrams, object diagrams, sequence diagrams, activity diagrams, component diagrams, and deployment diagrams. These diagrams provide information about the different components in detail.
5. **Chapter 5: Implementation and Testing**
   This chapter gives information about the tools used for implementing the projects as well as the testing carried out to increase the efficiency of the system.
6. **Chapter 6: Conclusion and Future Recommendations**
   This chapter concludes the report with the system's result and provides recommendations for future improvement.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Access to legal information, particularly constitutional law, has traditionally been challenging due to its complex language and structure. Individuals often rely on legal professionals or extensive document searches to find relevant legal information. However, Natural Language Processing (NLP) advancements, particularly Retrieval-Augmented Generation (RAG), have enabled AI-powered legal assistants to provide more efficient and context-aware responses to constitutional queries [3].

2.1.1 Challenges in Accessing Legal Information

Legal documents contain technical language, making it difficult for non-experts to understand them. Furthermore, conventional search engines that operate using keyword-based search often return irrelevant or incomplete results if the exact wording differs from the search query. In Nepal, language barriers and the lack of structured digital legal databases further hinder access to legal knowledge. Research has shown that AI-driven legal assistants can bridge this gap by offering automated and accessible legal support [4].

2.1.2 Retrieval Augmented Generation

The RAG (Retrieval-Augmented Generation) model is designed for text generation tasks, performing a process that involves retrieving information from given source data and utilizing that information to generate desired text. RAG model consists of the data processing pipeline for RAG usage, involving breaking down the original data into smaller chunks and converting text data into numerical vectors through embedding, which are then stored in a vector repository[5]

2.1.3 Vector Database

A vector database is a novel type of database developed to address the problem of long-term memory deficiency in LLMs. This database is specialized in efficiently storing and managing high-dimensional real-valued vector indices. Unlike traditional databases, a vector database represents queries in the form of real-valued vectors (embeddings) and supports a method to extract data with similar vectors. Therefore, it is optimized for storing and utilizing vectors obtained as results from AI models. [5]

2.1.4 Vector Search And Semantic Retrival AI

Traditional search methods rely on exact word matching, which is inefficient for legal research. Vector embedding, which represent text as numerical vectors, have transformed legal information retrieval by enabling semantic similarity-based searches. By applying cosine similarity, legal queries can be matched with relevant constitutional texts regardless of variations in wording [4]. Vector databases, such as FAISS and Pinecone, are widely used to store and retrieve vectored legal documents, significantly enhancing search accuracy and efficiency [6].

2.1.5 RAG For Constitutional Assistants

The RAG framework integrates retrieval-based search with AI-powered response generation. Unlike traditional virtual assistants that rely on predefined responses, RAG-based systems dynamically retrieve the most relevant constitutional provisions and generate structured, legally accurate responses. This approach ensures that users receive fact-based, context-aware answers instead of purely generated text without legal grounding [6].

2.1.6 LLM Training Stages

LLM training stages consists of following stages: First of all data is gathered(Data Corpus), data extraction, splitting chunks, embedding, vector store, prompt & search & generation
In Nepal, the need for such a solution is significant considering the complexities of the legal system. The purpose of LangChain based application "कानुन" aims to address this necessity by leveraging cutting-edge AI technology and natural language processing. "कानुन" intends to be a dependable and user-friendly platform for answering constitutional related questions, warning citizens, legal experts and government representatives alike.

2.2 Literature Review

The LLMs are utilized for the development of applications and their capabilities in a range of tasks, with OpenAI's ChatGPT being popular among millions of users. Large Language Models (LLMs) are used as a type of model in LangChain to accept a text string (prompt) and output a text string. As a stateless model, each transaction to the LLMs API endpoint is independent. Vector stores and retrievers can take multiple forms as memory insertion. The use case of common end-to-end applications includes autonomous agents, chatbots,

code understanding agents, extraction question answering over documents, and analyzing structured data [8].

The recent advancements in generative AI technology addresses the challenge of information scarcity and proposes remedies by harnessing the LLMs capabilities. There is liberal framework for implementing generative AI service by effectively combining and orchestrating various technologies. The implementation procedure of the RAG model includes source data collection and extraction, chunk generation, embedding, building the vector database, integration of prompt and search results. The LLM and RAG models requires reducing of the size and complexity of LLM models to achieve effective results with smaller and simpler models. The combined efforts of these initiatives are expected to expand the utilization of generative AI technology which will be practical in the real world[9].

The large language models (LLMs) plays important role in providing substantial assistance to various stakeholders, including decision-makers, administrators, and citizens. The working mechanisms of legal assistant tailored for interacting with legal resources can be observed in the regulatory activities of major industrial powers. An unexpected outcome of scaling LLMs has been a qualitative shift in their behavior, affecting their capabilities. Furthermore, developing and testing the LLM-based system with complex legal questions remains a challenge. The authors highlight the future potential of these systems in effectively performing legal tasks[10].

Large Language Models(LLMs) are evaluated for their efficiency amd repeatability in automating question-answering tasks across multiple documents. Prompt formatting may always not meet the satisfactory criteria. In order to improve the performance of LLM, optimization of embeddings and vector database can be done. Additionally, using OpenAI models via API calls helps reduce response times when upgrading local infrastructure is not feasible. The focus is on leveraging AI to improve human efficiency rather than merely validating its output[11].

A RAG-based system is developed in order to address two key challenges in the legal domain: the risk of generating inaccurate content or opacity of AI-driven tools. The framework assists users in querying legal documents and providing reliable answers. The cleaned legal dataset with expert-verified questions, answers and, references is used for

benchmarking the created system. The system is made publicly accessible, mainly for benefiting law students and legal professionals[12].

A RAG-based method is mainly used in the field of professional knowledge-based question answering, where model companies, embedding, and chat API interfaces and frameworks like LangChain have already integrated the RAG process. There are limitations in using the RAG approach, such as ranking issues, token limits, and fine segmentation drawbacks, which affect information retrieval. The effect of PDF recognition on RAG through various settings, data preparation, and extractive question results can be measured. Comparisons with deep learning-based document parsing methods provide a deeper understanding of the relationship between RAG quality and document parsing accuracy[13].

Traditional journalism has been highly affected by the growing LLM usage. Around 17,000 jobs lost in the past eight years and the market value of newspapers are decreasing. the common objection that lawyers have always adapted to technological changes, and these changes will lead to legal action regarding the proper scope and demand for legal services. Regulatory barriers may not prevent machine intelligence from challenging the lawyers' monopoly, but restrictions on the unauthorized practice of law create obstacles for AI in providing legal services to non-lawyers. Ultimately, the rise of machine intelligence is expected to make legal services more transparent and affordable[14].

The IQLS framework mainly focuses on optimizing models to fulfill this task on given required data in a structured format. Retrieving, consolidating and applying the right data is therefore a huge task. The IQLS system is designed to handle various complex and diverse queries with ease. Its ability to handle language ambiguity is one of its key features which enhances user experience by accurately. The IQLS extends the capabilities of LLMs to advanced tasks like route planning while utilizing available data structures. The adaptability of the models depends on data mapping, where the schema influences query results. This study provides practical testing in real-world environments, and future research could focus on developing more error-proof methods for parsing complex structured outputs from LLMs to enable broader applications[15].

A legal judgment prediction system is designed to predict and explain court decisions as cases are adjudicated. The study enhances prediction accuracy by applying grounded legal principles. It utilizes Hierarchical BiLSTM-CRF classifiers to extract relevant sentences based on rhetorical roles in legal judgments. The model processes inputs in segments of

2048 words, generating predictions efficiently within a token limit of 4096. A key limitation is its exclusive focus on Supreme Court of India judgments, which may reduce its applicability to other courts or jurisdictions. The study also explores the effectiveness of various LLMs and transformer architectures using the ILDC-multi dataset[16].

The Legal Reasoner framework enhances legal judgment prediction by integrating legal knowledge infusion, case-law retrieval, multi-hop reasoning, judgment synthesis, training, and optimization. It improves upon state-of-the-art models, achieving a 7.8% average accuracy increase across datasets. The system predicts legal case outcomes based on facts, relevant statutes, and historical precedents. By combining multiple innovative components, the framework enhances AI-assisted legal decision-making, improving efficiency and consistency in legal processes[17].

AI and machine learning are increasingly shaping the legal field, improving human capabilities and automating legal processes. The methodology involves analyzing legal information based on semantics, making legal predictions, and assisting judicial automation to improve efficiency at a lower cost. The paper estimates that within the next 20 years, over 100,000 legal jobs could be automated, with AI handling 22% of a lawyer's tasks and 35% of a clerk's duties. The study emphasizes the need for legal professionals to embrace AI-driven transformations.[18]

AI and machine learning are revolutionizing the legal field by enhancing human capabilities and streamlining legal processes. This study explores the latest advancements and emerging research trends in AI-driven legal applications. The methodology involves analyzing legal information based on semantics, making legal predictions, and assisting judicial automation to improve efficiency at a lower cost. The paper estimates that within the next 20 years, over 100,000 legal jobs could be automated, with AI handling 22% of a lawyer's tasks and 35% of a clerk's duties. The study emphasizes the need for legal professionals to embrace AI-driven transformations.[19]

The impact of AI-produced legal documents is transforming how legal experts and students assess the credibility and standards of written content. This study examines the preferences of legal professionals when analyzing documents they perceive as either AI-generated or human-written. 75 legal professionals and law students were asked to assess the precision and linguistic quality of legal documents. The findings reveal a clear preference for materials believed to be authored by humans. The study emphasizes the necessity of

thoughtfully incorporating AI into legal writing and calls for additional research into how legal practices should adapt to the rise of AI-generated content.[20]

In a research, the benchmark designed to evaluate the retrieval step of RAG pipelines within the legal space. It serves as a critical tool for companies and researchers focused on enhancing the accuracy and performance of RAG systems in the legal domain. In this paper methodology is used to extract highly relevant text snippets from legal documents rather than retrieving entire documents and retracing the context used in LegalBench queries back to their original location in a legal corpus. The limitation or research gaps in the research manuscript is focusing on the four sources of dataset include NDAs, M&A agreements, various commercial contracts and privacy policies of consumer-facing online companies. The creation of both LegalBench-RAG and its lightweight counterpart and LegalBench-RAG-mini addresses a critical gap in the evaluation of RAG system particularly in contexts where legal accuracy is paramount.[21]

Open-source models are transforming legal language modeling and analysis, offering efficient ways to process complex legal texts. This study explores the use of open-source models from Hugging Face to analyze legal documents, with a focus on the Indian Constitution. The methodology involves preprocessing constitutional data, segmenting it into smaller chunks, and utilizing Milvus/Pinecone and LangChain Vector Stores for data storage and management. This approach streamlines model creation while ensuring that the resulting language models are specifically adapted to the nuances of legal texts. The study also emphasizes privacy protection through data access control, minimal data exposure, and limited dependencies. By applying these techniques, the research highlights how legal professionals and policymakers can better navigate and interpret extensive legal texts.[22].

Although chat bots have existed for decades, the emergence of transformer-based large language models (LLMs) has captivated the world through the most recent wave of artificial intelligence chat bots, including ChatGPT. Transformers are a type of neural network architecture that enables better contextual understanding of language and efficient training on massive amounts of unlabeled data, such as unstructured text from the internet. As LLMs have increased in size, their improved performance and emergent abilities have revolutionized natural language processing. Since language is integral to human thought, applications based on LLMs have transformative potential in many industries. In fact, LLM-based chatbots have demonstrated human-level performance on many professional

benchmarks, including in radiology. LLMs offer numerous clinical and research applications in radiology, several of which have been explored in the literature with encouraging results. Multimodal LLMs can simultaneously interpret text and images to generate reports, closely mimicking current diagnostic pathways in radiology. Thus, from requisition to report, LLMs have the opportunity to positively impact nearly every step of the radiology journey. Yet, these impressive models are not without limitations. This article reviews the limitations of LLMs and mitigation strategies, as well as potential uses of LLMs, including multimodal models. Also reviewed are existing LLM-based applications that can enhance efficiency in supervised settings.

In conclusion, LLM integration into the legal domain holds great promise for democratizing the access to legal knowledge all over the glove. However, the complexity and interpretation ability might be the factor to choose the correct requirements.

CHAPTER 3: SYSTEM ANALYSIS

3.1 System Analysis

System analysis is a problem-solving method that involves examining the broader system, deconstructing its components, and understanding how it operates to achieve specific goals. The first step in addressing a system-related problem is to analyze the system itself [23]. This process includes breaking down the system into its constituent parts and evaluating how these parts function collectively. In some cases, understanding a system may involve disabling certain components to observe the impact or modifying parts to analyze the resulting changes.

System analysis is conducted to study a system or its components to identify its objectives. It focuses on improving application performance and ensures all components work efficiently to achieve their goals[24]. It includes analyzing business processes, gathering operational data, understanding the flow of information and developing solutions to overcome system weaknesses. It supports the achievement of organizational goals.

Furthermore, system analysis includes subdividing complex processes involving the entire system, identifying data stores, and analyzing manual processes. According to the *Merriam-Webster Dictionary*, system analysis is defined as “the process of studying a procedure or business in order to identify its goals and purposes and create systems and procedures that will achieve them in an efficient way” [25].

3.1.1 Requirement Analysis

Requirement analysis is a critical phase in system development that require identifying, documenting, and understanding the requirement and expectations of stakeholders to ensure the application achieves its goals. This phase focuses on figuring out what the system should do, the constraints it must operate within, and how it will interact with other systems [26].Requirements analysis involves various tasks that help engineers understand stakeholder demands and explain them in simple and visual ways. It is essential to a software or system project's success.

For a project to be successful, its requirements must be testable, actionable, documented, measurable, traceable. Requirements analysis involves various stakeholders, such as project sponsors, throughout the project as well as end users whose inputs are most

important. The best results typically occur when all parties work together to develop a high-quality requirements document .

There are many requirements of the system which should be analyzed so that effective output can be produced with available resources. Following requirement analysis is carried out for the effectiveness of our system.

i. Functional Requirements

Functional requirements describe the features and functions that must be implemented to accomplish the system's requirements. It's important to make them sort out both for the development team and the stakeholders.[27]

Functional Requirements help as a comprehensive and detailed set of specifications, outlining the specific tasks and functionalities that a software system or its components must perform. These requirements define how the system should behave, including how it processes inputs, generates output, and handles the important steps for specific functions. The main goal of functional requirement is to lead the design and development of the software, ensuring it meets the requirement of its users

Functional requirement gathers and analyzes the functional part of the system. In 'The LangChain Based Chat Application System', users are allowed to view laws and regulations or sign up and log in the dashboard. The users are allowed to search for constitutional queries. The users are allowed to give the text as an input to the system. Once the input is passed, the given text is then processed through various NLP techniques. The system analyzes the input into relevant articles and clauses from the Constitution of Nepal. Functional requirement of the system can be described with the help of use case diagram as following:

Use Case Diagram

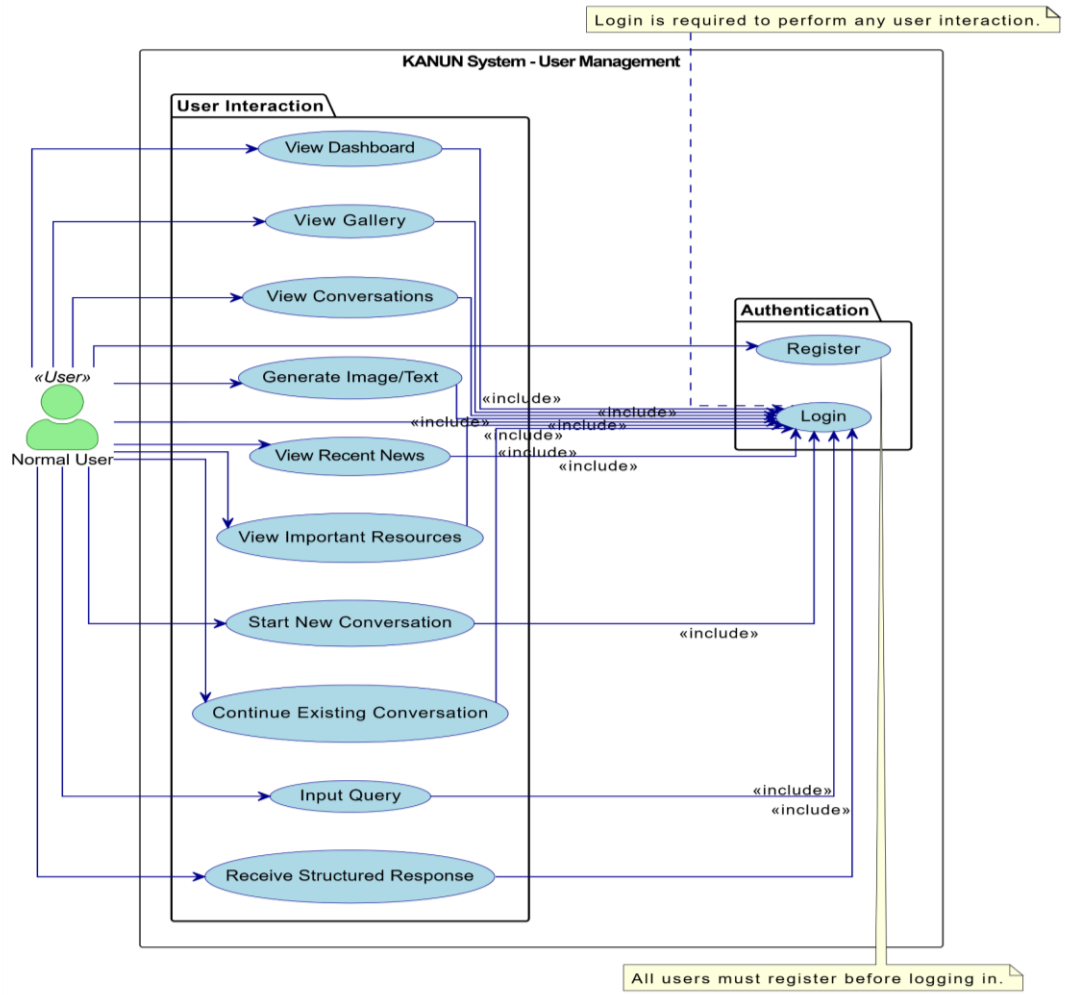


Figure 2: Use Case Diagram for Normal User

Above use case diagram illustrates the interaction between normal user and the system. Here, the normal user is able to view dash board, view Gallery, View Conversations, View recent news, etc. after login and registration

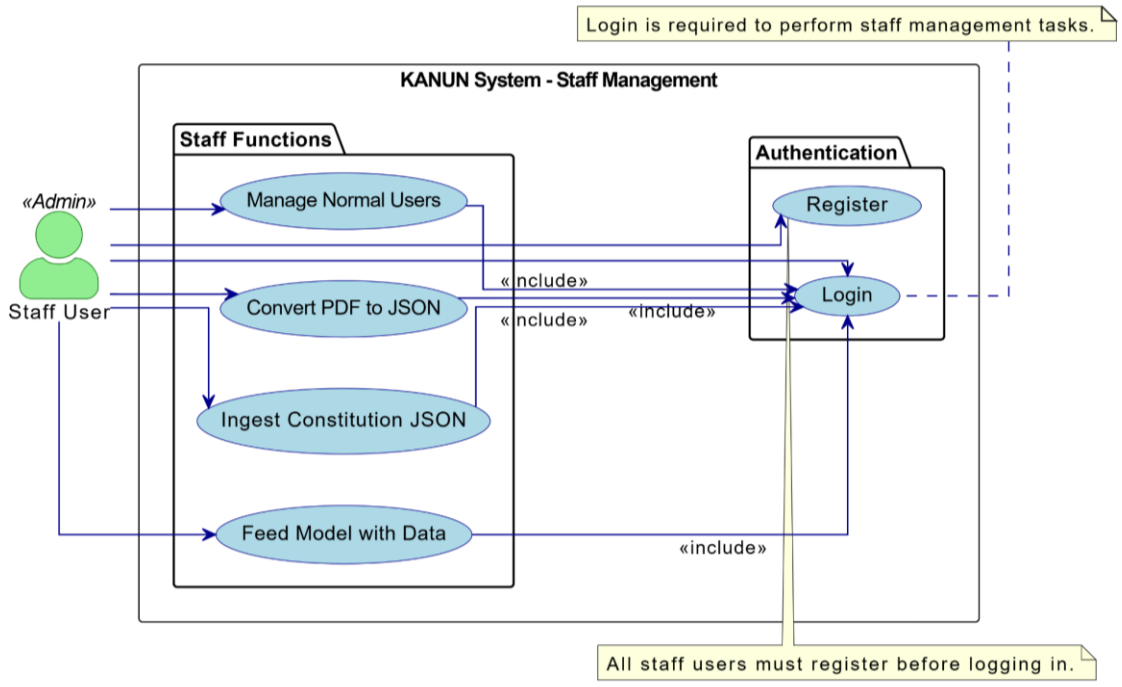


Figure 3: Use Case Diagram for Staff Management

Above use case diagram illustrates the interaction between staff user and the system. Here, the staff user is able to view manage normal user, convert PDF to JSON, Injest Constitution, and Feed model after Login .

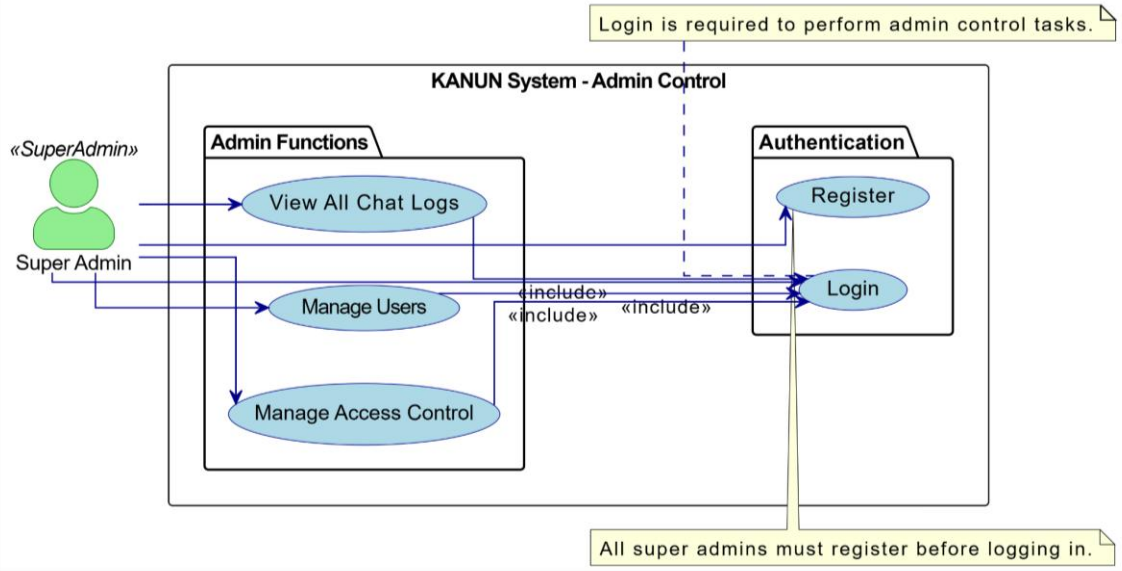


Figure 4: Use Case Diagram For Admin Control

Above use case diagram illustrates the interaction between Super admin user and the system. Here, the Super Admin is able to view all chat logs, manage users, Manage access control.

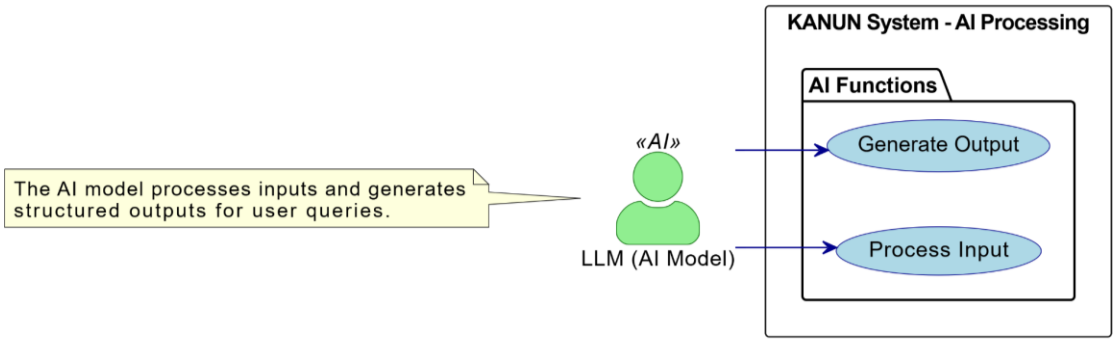


Figure 5: Use Case diagram for AI Processing

Above use case diagram shows the LLM interacting with the system

Use-case Table

Table 1: Use Case Login Table

Fields	Details
Use Case	Login
Actor(s)	Normal User, Staff User, Admin
Description	This use case allows users to log in to the system using their credentials
Priority	High
Status	Implemented
Pre-Condition	The user must have valid login credentials stored in the system
Post-Condition	The user is authenticated and redirected to their respective dashboard
Main Flow	The user enters their credentials, submits the login form, and is authenticated by the system
Alternate Flow	If the user enters incorrect credentials, an error message is displayed, and they can retry
Exceptional Flow	If the system encounters an error (e.g- server downtime), an error message is displayed.

Table 2: Use Case Register Table

Fields	Details
Use Case	Register
Actor(s)	Normal User
Description	This use case allows users to register for an account in the system.
Priority	Medium
Status	Implemented
Pre-Condition	The system must allow registration and the user must provide valid details(e.g- email, password).
Post-Condition	The new user account is created and stored in the system and users can login.
Main Flow	The user fills out the registration form with required details and submits it. The system validates the input and creates the account.
Alternate Flow	If the user provides incomplete or invalid details, the system displays an error message.
Exceptional Flow	If account creation fails due to a system error, the user is prompted to retry later.

Table 3: Use Case Dashboard Table

Fields	Details
Use Case	Dashboard
Actor(s)	Normal User, Staff User
Description	This use case allows users to view their personal dashboard which includes token usage, recent news, quiz game and resources.
Priority	Medium
Status	Implemented

Pre-Condition	The user must have register credentials with valid user id and password.
Post-Condition	The user can perform necessary action in the system after successful login.
Main Flow	The user navigates to the dashboard and the system displays the dashboard features and content.
Alternate Flow	If no data is available then the system displays a default message.
Exceptional Flow	If the dashboard fails to load then the login failed message has to be displayed.

Table 4: Use Case Generate Text/Image Table

Fields	**Details**
Use Case	Generate Text/Image
Actor(s)	Normal User
Description	This use case allows users to generate text or images using the LLM.
Priority	Medium
Status	Implemented
Pre-Condition	The user must be logged in.
Post-Condition	The generated text/image is displayed to the users.
Main Flow	The user inputs a prompt,submits it and the LLM generates the output.
Alternate Flow	If the user input is invalid then the system displays an error message.
Exceptional Flow	If generation fails then the system logs the error and displays a message.

Table 5: Use Case Convert PDF to JSON Table

Fields	Details
Use Case	Convert PDF to JSON
Actor(s)	Staff User
Description	This use case allows staff users to convert PDF files into JSON format for ingestion.
Priority	Medium
Status	Implemented
Pre-Condition	The PDF files must be uploaded to the system.
Post-Condition	The PDF is converted into JSON format and stored into the system.
Main Flow	The staff user uploads the PDF and the system convert it into JSON.
Alternate Flow	If the PDF is invalid, the system displays an error message.
Exceptional Flow	If conversion fails, the system logs the error and displays a message.

Table 6: Use Case User Management Table

Fields	Details
Use Case	User Management
Actor(s)	Admin
Description	This use case allows super users to manage user accounts, including adding, editing and deleting users and assigning roles.
Priority	High
Status	Implemented
Pre-Condition	The actor must be logged in with appropriate permissions.

Post-Condition	User accounts are updated, added or removed from the system.
Main Flow	The actor navigates to 'User Management', performs an action and confirms the update.
Alternate Flow	If the action is canceled, no changes are made.
Exceptional Flow	If an account management fails, the system logs the error and displays a message.

Table 7: Use Case Logout Table

Fields	Details
Use Case	Logout
Actor(s)	Normal User, Staff User, Admin
Description	This use case allows users to log out of the system.
Priority	High
Status	Implemented
Pre-Condition	The user must be logged in.
Post-Condition	The user session is terminated and they are redirected to the login page.
Main Flow	The user clicks the logout button and the system terminates their session.
Alternate Flow	If the logout action is canceled then the session remains active.
Exceptional Flow	If logout fails due to a system error, the session remains active and the system logs the error.

ii.Non Functional Requirements

The non-functional requirements define the attributes or qualities that a system or software should have, in addition to what it does. It is concerned with the performance, usability, and other additional qualities of the system. The non-functional requirements often apply to the system as a whole.[28]

The non-functional requirements of this RAG-based application are as follows:

a) **Performance**: Since backend handles all the complex tasks, users will be able to access the query application easily and fast with minimum hardware and average internet connection using the frontend.
b) **Look and feel**: The user interface is easy to use and light colors are used with beautiful design to enhance usability.
c) **Security**: Users must login with valid credentials to access advanced features. Their credentials are stored securely into a database using robust hashing techniques to ensure data privacy and security.
d) **Reliability**: The constitutional articles and clauses are stored in the database for quick retrieval. So, every time a user met with accurate and relevant information without delays making the system highly dependable.

3.1.2 Feasibility Study

A feasibility study evaluates the project whether it is practical and achievable in terms of technical, economic, legal, operational and time feasibility factors. The study is also intended to identify possible challenges and problems that we may encounter while completing the project.[29]

Feasibility study helped the authors to determine the strength and weakness of our system. The contents and recommendation of this feasibility study helped us a sound basis for determining how to proceed further. It also helped in making decisions regarding which software to use, hardware combination etc. Some feasibility studies carried out are as follows.

i. Technical Feasibility

Technically, the system is a RAG-application which is developed to solve Constitutional Queries using LLM, RAG, vector stores and Django Python as core programming language. The backend will use SQLite3 for database management. The frontend is written using HTML, CSS and Bootstrap. The technologies that are needed for the development of the application.

ii. Operational Feasibility

The operational feasibility of the "कानुन" (KANUN) application relies on several key factors. A strong foundation requires robust infrastructure, such as servers, compatible

software (like the Django framework, a language model, and possibly a vector database), and reliable internet. Skilled human resources are important like technical staff and legal experts for accurate content and smooth operation. Operational processes, including data updates, system maintenance, and user support, must be well-defined and consistently executed. Security and privacy are critical with strong measures and regulatory compliance ensuring users trust. Scalability and a sustainable funding model are essential for the application's long-term success. A detailed assessment of resources, costs and risks is needed before starting successful implementation. Regular monitoring will help to improve the application and keep it effective over time.

iii. Economic Feasibility

The economic feasibility of "कानुन" considers the cost associated with developing, implementing, and maintaining the RAG-based application, as well as the potential returns and savings it can generate. Preprocessing and classifying textual documents for performing RAG operations on the Constitution of Nepal is a task that modern hardware can efficiently handle. The system is lightweight and doesn't require much bandwidth so low-cost servers can also be used. So, this system is economically feasible.

iv. Schedule Feasibility

This system has been developed to partially fulfill requirements of BSc. CSIT course. We had 3-4 months to complete the system. The lifecycle of the system can be executed as shown in the schedule below.

Table 8: Time Schedule

Name	Begin Date	End Date	Duration (Days)
Brainstorming	09/25/2024	10/02/2024	8
Project Planning	10/03/2024	10/12/2024	10
Requirement Analysis	10/13/2024	10/19/2024	7
Collecting Resources	10/20/2024	10/28/2024	9
Developing the System	10/29/2024	01/15/2025	79
Testing	01/16/2025	02/10/2025	26
Documentation	09/25/2024	02/28/2025	157

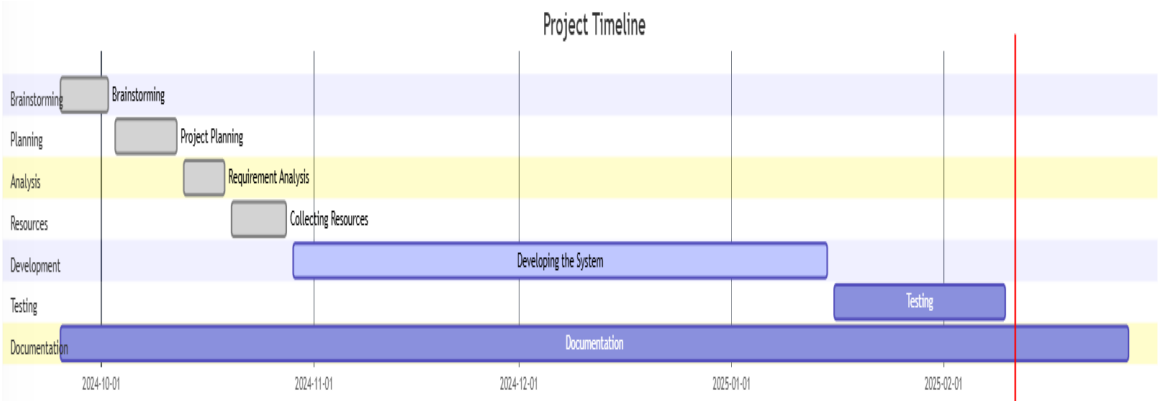


Figure 6: Gantt Chart of system development schedule

3.1.3 ANALYSIS

3.1.3.1 Class Diagram

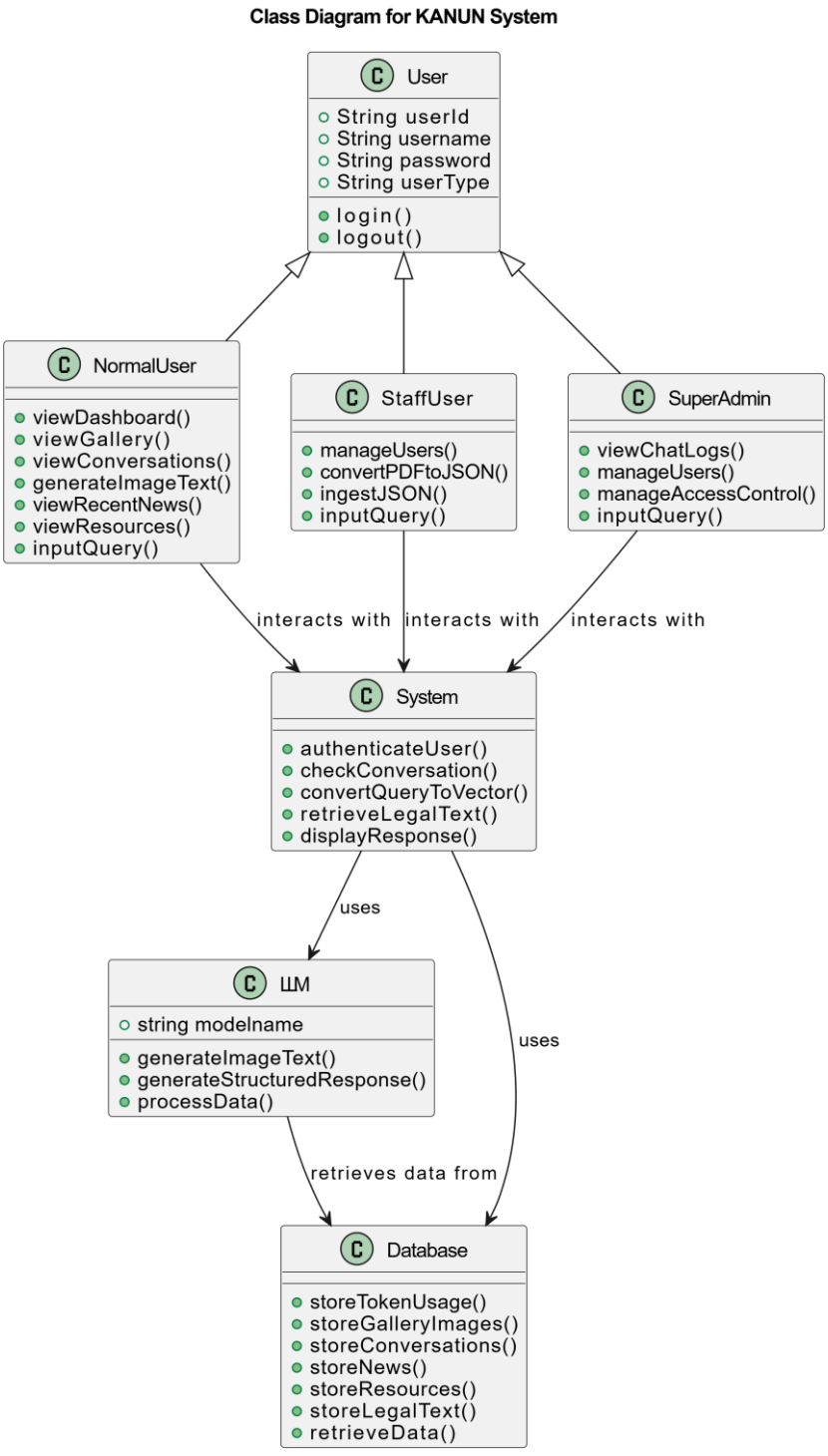


Figure 7: Object Modeling using Class Diagram for Kanun

The above diagram illustrates Class diagram of the Kanun Application. The Kanun applications mainly consists of User, System LLM and database and main classes. The

reader will get an in-depth understanding of the object model and its elements. The basic factors of an object model are classes and objects. An object is a physical component in the object-oriented domain which may be tangible such as a person, car or intangible such as a project. A class is a representation of objects, It represents a group of objects that have similar properties and exhibit an expected behavior. [30]

3.1.3.2 Dynamic Modeling

System diagrams are tools that help you to understand how complex systems work. They're particularly helpful for showing you how a change in one factor may impact elsewhere. They're excellent tools for flushing out the long-term impacts of a change. Importantly, a good systems diagram will show how changing a factor may affect itself. Drawing a systems diagram is a good way of starting to design a computer model. The technique helps you to plan the structure of the system as per required to model. It highlights the factors and their relationships that are important, and helps you to start measuring the linkages between factors The sequence diagram below shows the sequence in which objects in the system operate with one another.

A normal user visits the dashboard by entering a query or interacting with the system. The query is sent to the backend which is processed through LangChain and RAG pipeline. Relevant articles and clauses are fetched from the Constitution Database and contextual data is retrieved from the Vector Store. The processed response is then displayed to the users. Users can also upload raw PDF files containing constitutional text which are ingested and processed into the Vector Store for future queries or conversation.[31]

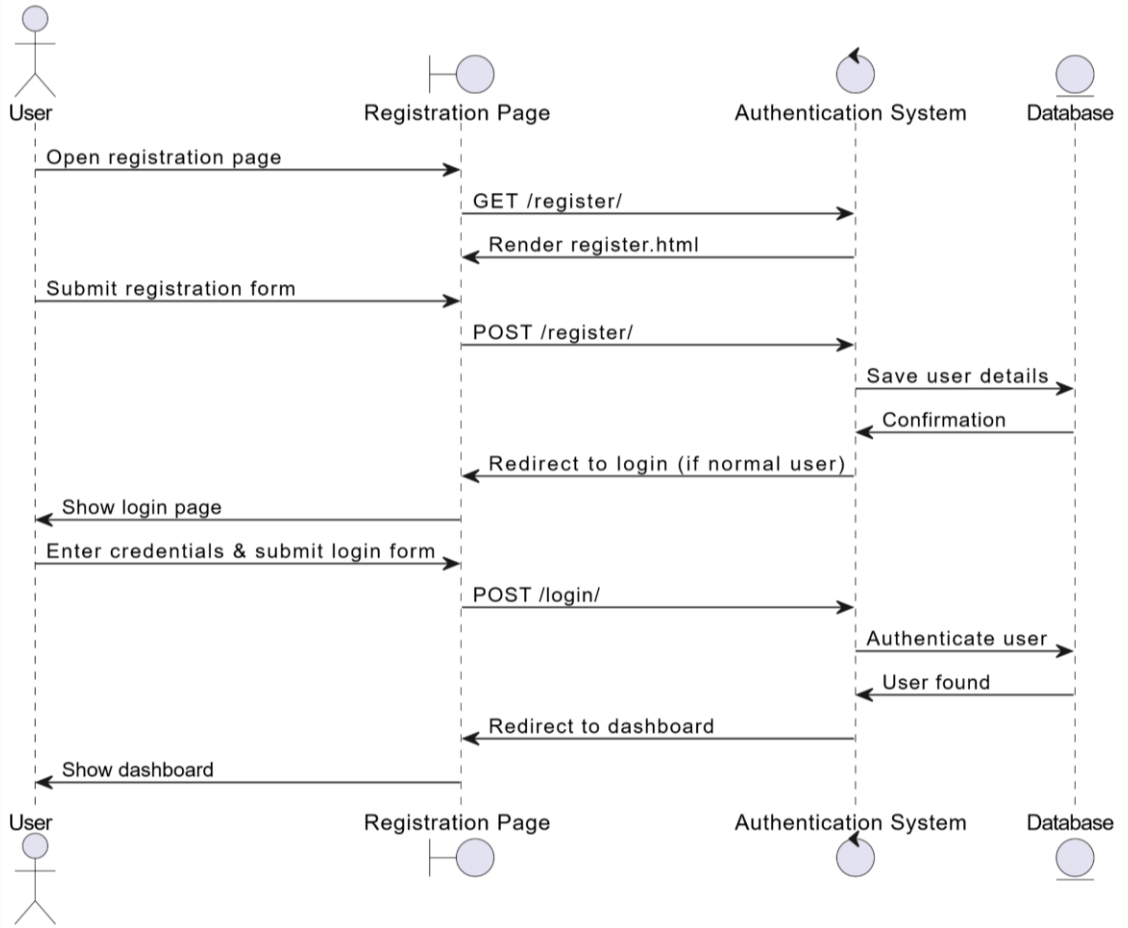


Figure 8: Dynamic modeling of Normal User Authentication Flow using sequence diagram

The above sequence diagram illustrates the interaction between User, Registration, Authentication and database. The process begins when the user opens the registration page and ends with the user being redirected to their dashboard. The process begins when the User accesses the registration page through the Browser. Browser sends request to the system. The system generates the registration form and sends it back to the Browser for displaying. The User fills out the form and submits it. The system processes this information, saving it in the Database , which confirms the operation. Afterward, the system redirects the Browser to the login page. The User enters their credentials and submits the form, triggering a request. The system verifies the credentials against the Database , and upon successful authentication, redirects the Browser to the dashboard, completing the flow.

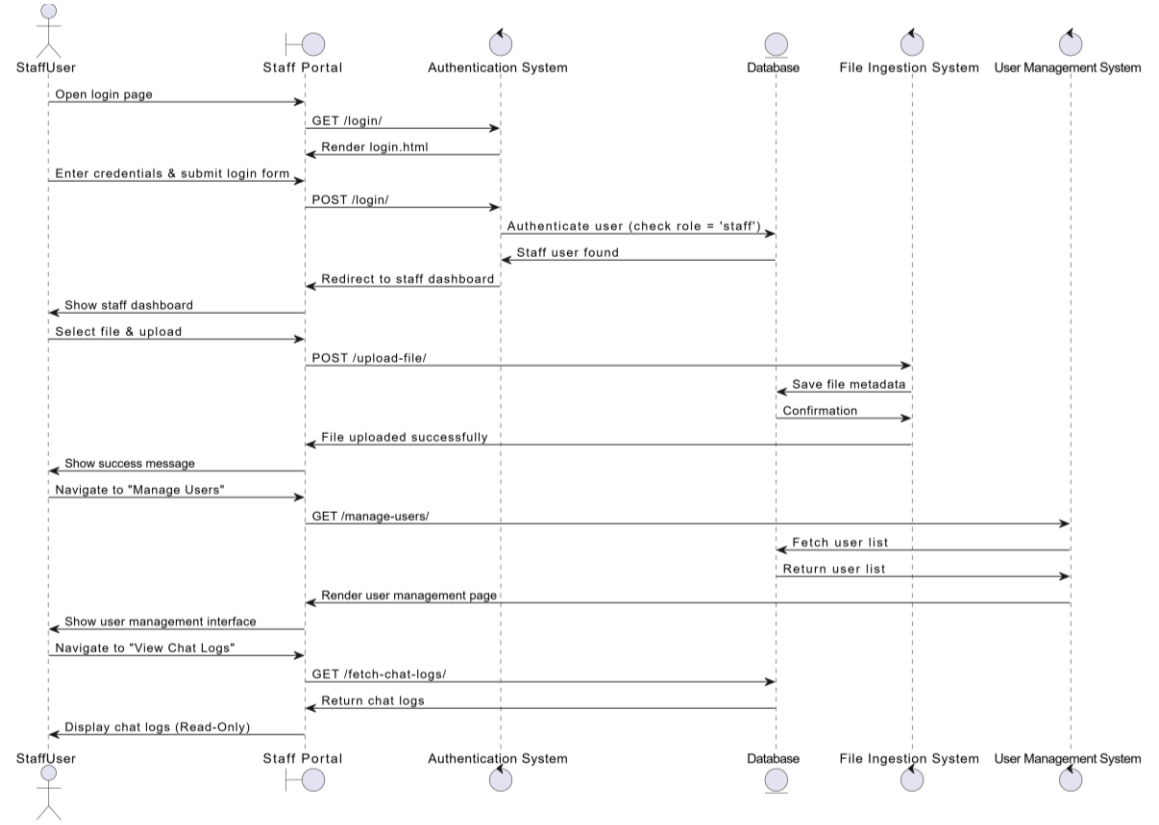


Figure 9: Dynamic modeling of Staff User Workflow using sequence diagram

Using the Authentication System and Database, the Staff User initiates authentication. When they enter the Staff Portal after successfully logging in, they are taken to the dashboard, where they may monitor chat logs, manage users, or upload files. While user management retrieves and updates information, file uploads store metadata in the database. It is possible to view, edit, and save chat logs back into the system. This process guarantees smooth staff operations for administrative, data management, and authentication duties.

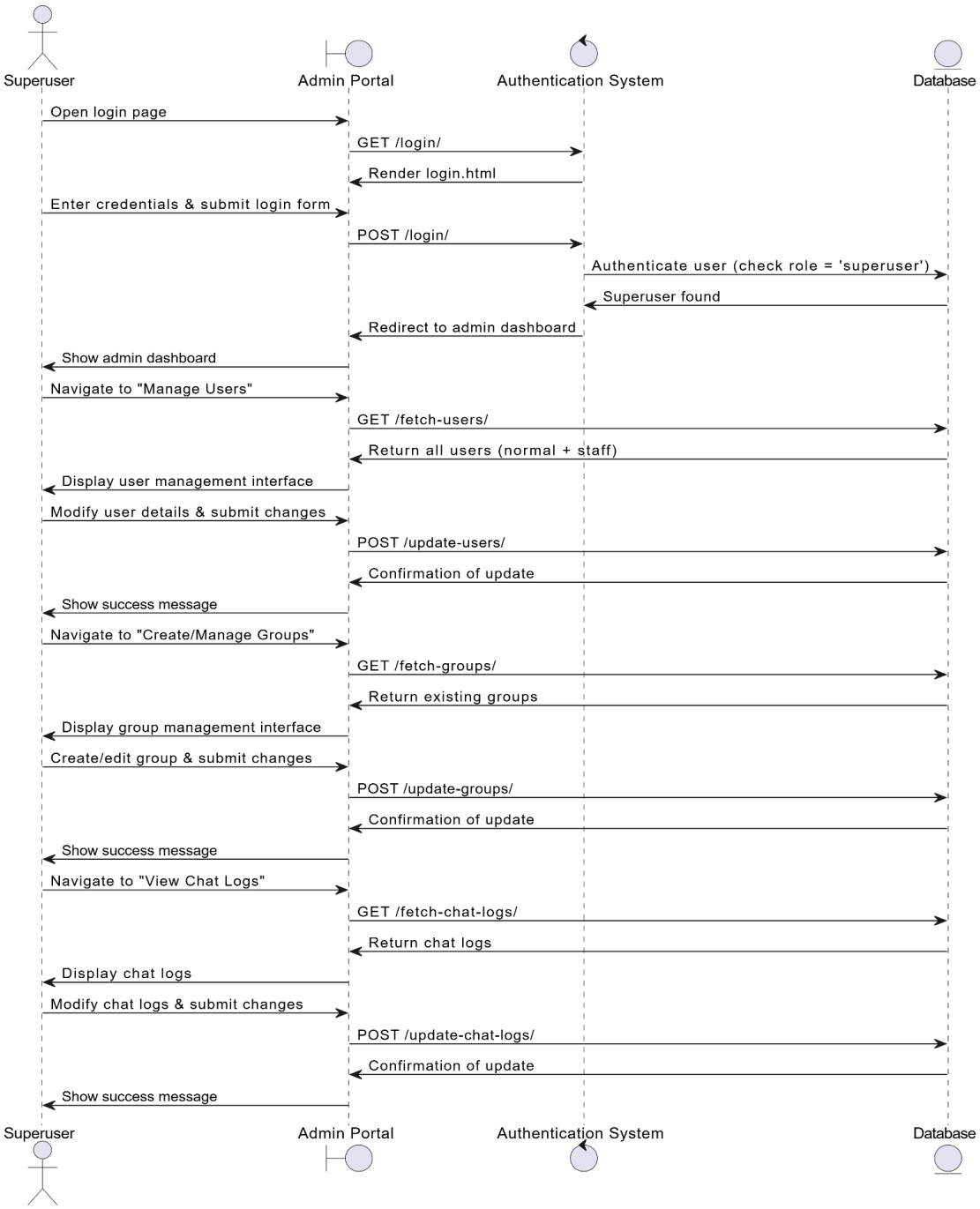


Figure 10: Dynamic modeling of Super User Workflow using sequence diagram

The Superuser Workflow begins when the Superuser accesses the login page via the Admin Portal , prompting the system to retrieve and display the login form. The Superuser enters their credentials and submits the form, which the system verifies against the Database , ensuring the user has the "superuser" role. Upon successful authentication, the system redirects the Superuser to the admin dashboard. From there, the Superuser can manage all users, including normal users and staff users, by navigating to the "Manage Users" section. The system fetches the complete user list from the Database and displays it for

modification. After making changes, the Superuser submits the updates, which are saved to the Database , and the system confirms the operation with a success message. Optionally, the Superuser can create or manage groups by accessing the "Create/Manage Groups" section, where the system retrieves existing groups from the Database and allows for creation or editing. Similarly, the Superuser can view chat logs by navigating to the "View Chat Logs" section, where the system retrieves and displays logs. If needed, the Superuser can also modify chat logs, submitting updates that are saved to the Database , with the system confirming the changes. This workflow ensures seamless interaction for the Superuser , enabling comprehensive management of users, groups, and chat logs while providing clear feedback at each step.

3.1.3.3 Process Modeling

Process modeling is the representation of business processes or workflows. A process model allows visualization of business processes which helps organizations see and understand their internal business procedures by making it easier to be managed and improve efficiently [32].

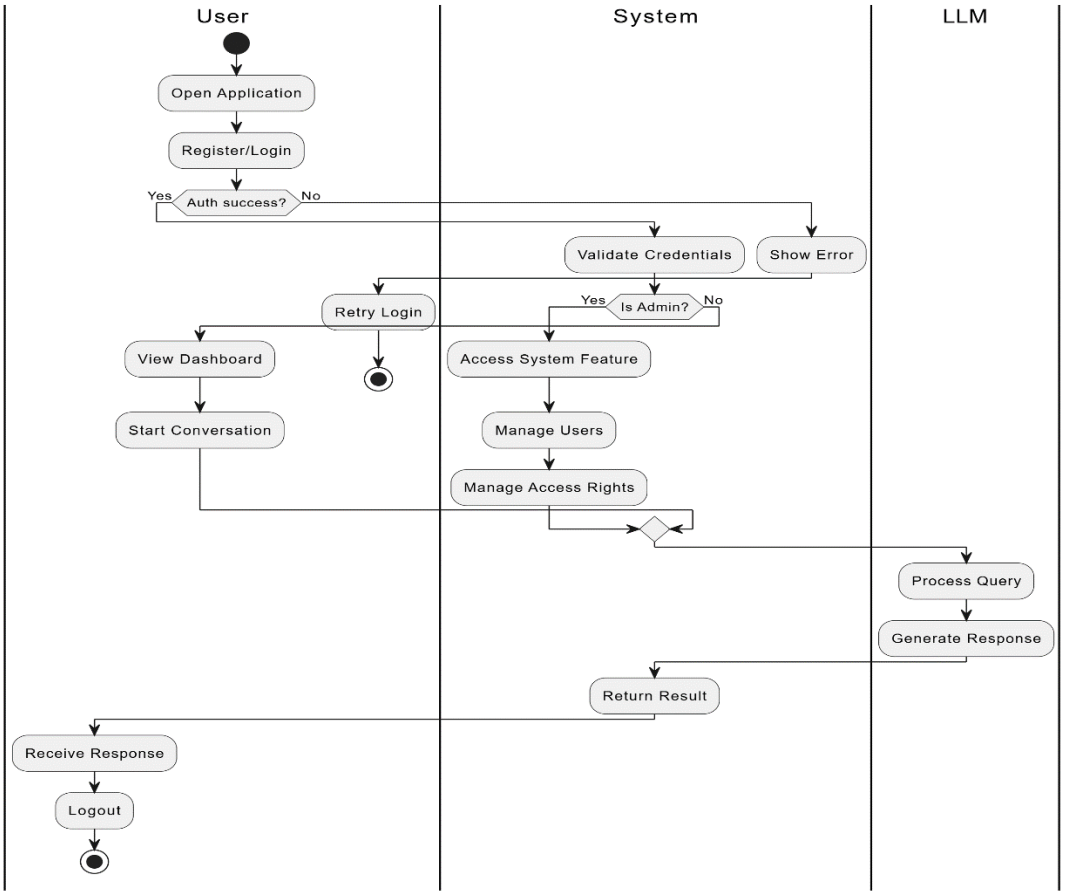


Figure 11: Activity Diagram of Kanun Application

The above activity diagram illustrates the workflow of the Kanun application performing the RAG operation on Constitution of Nepal across the three different swimlanes: User, System, and LLM (Large Language Model). The process begins with the user opening the application and attempting to log in. If the credentials were invalid, the system displays an error and suggesting user to login again. Otherwise, authentication proceeds. The user can view the dashboard and start a conversation. If the user was an admin then they can manage users and permissions, with the system by handling these tasks. Non-admin users directly moves to submits queries. When a user process a query or start conversation, the system forwards it to the LLM for processing.

The system interacts with an LLM for handling user queries. LLM generates a response which was send back to the system and response was displayed to the users. Finally, the process concludes when the user logout. Throughout the workflow, decision points ensure that users only access features appropriate to their roles. The integration of LLM model enhances the system's ability to process complex queries efficiently, creating smooth interaction between the user, system, and LLM.

CHAPTER 4: SYSTEM DESIGN

4.1 System Design

4.1.1. Refinement of Activity diagrams

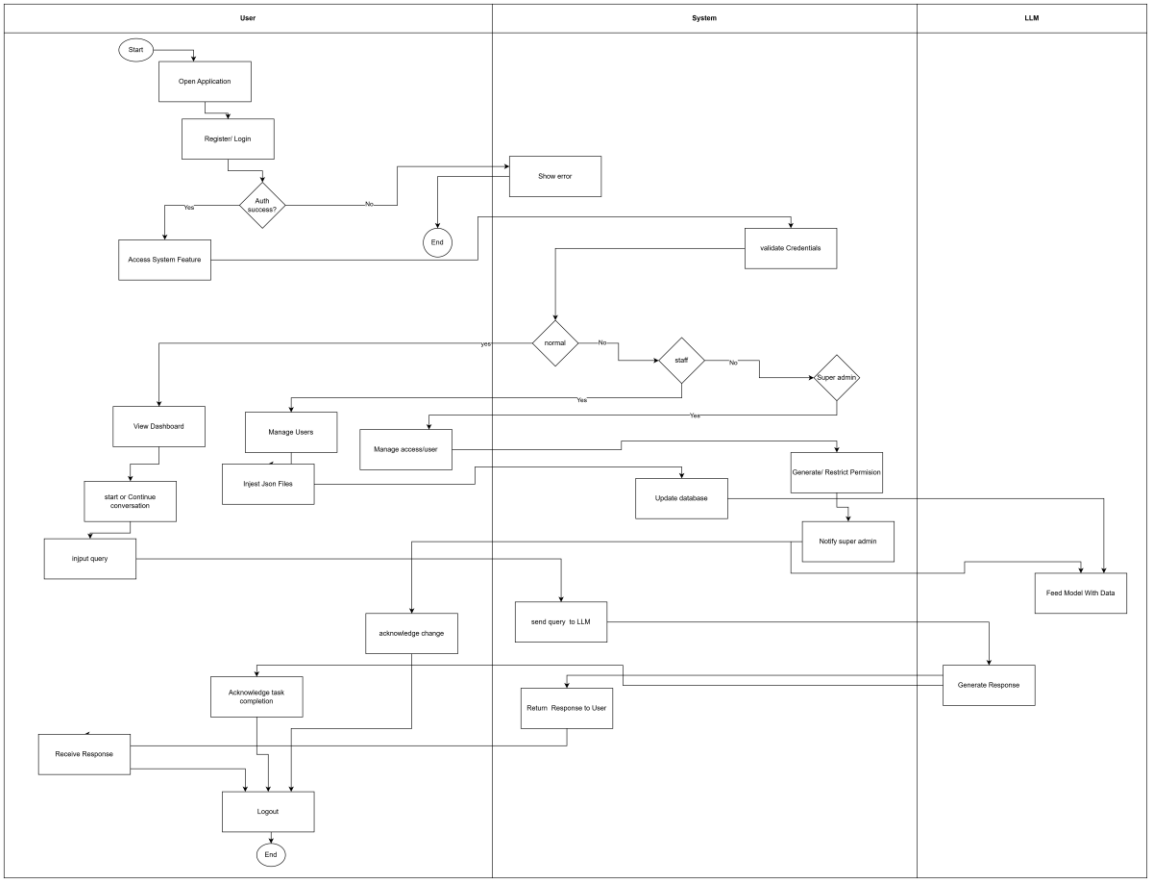


Figure 12: Refinement of Activity Diagram

The above figure illustrates the refinement of the activity diagram that gives information about the proper workflow of the system. It begins with the user opening the application and attempting to log in. If the credentials are invalid, the system displays an error; otherwise, authentication proceeds. Once logged in, the user can access system features, and the system validates credentials before granting access. The user can then manage data, view the dashboard, and input information, which the system processes and updates in the database. At this stage, a feature extraction mechanism may be triggered to analyze and process data.

The system also interacts with an LLM for handling user queries. When a user submits a query, the system forwards it to the LLM, which retrieves relevant data and generates a response. The system returns this response to the user. The user can acknowledge changes

or submissions. Finally, the process concludes when the user logs out. Throughout the workflow, decision points ensure that only valid inputs proceed, and data handling is structured. The integration of an AI model enhances the system's ability to process complex queries efficiently, creating smooth interaction between the user, system, and LLM.

4.1.1 Component Diagram

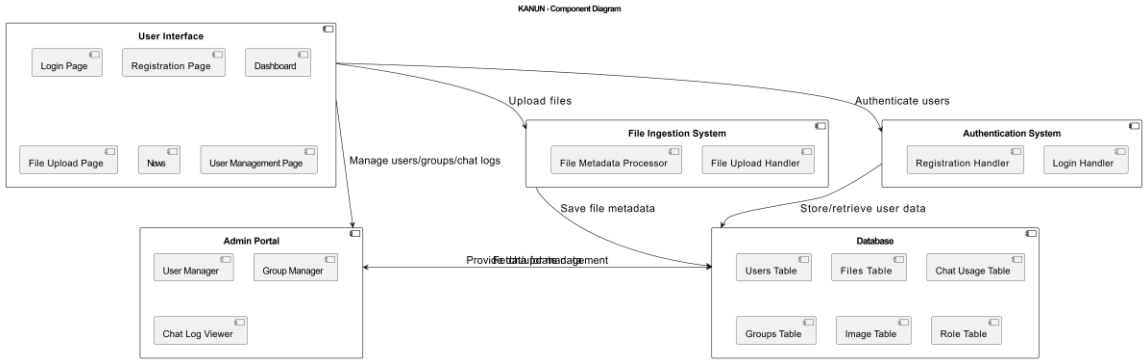


Figure 13: Use Case diagram for AI Processing

The above component diagram illustrates the high level design of the system and the relationship between the components. For this system, the components include User interface, Admin portal, File Ingestion System, Authentication and Database. The user interface components with sub-components like login page, register page, dashboard, file upload page, news, user management page, which provides the file uploading features, display latest news and maintains the chat history. The Authentication System components contain registration handler and login handler as sub-components which are responsible for validate credentials before granting access. The File ingestion System contains File metadata processor and file upload handler that save files metadata and manages to upload the raw data files which converted into JSON files.

4.1.2 Deployment Diagram

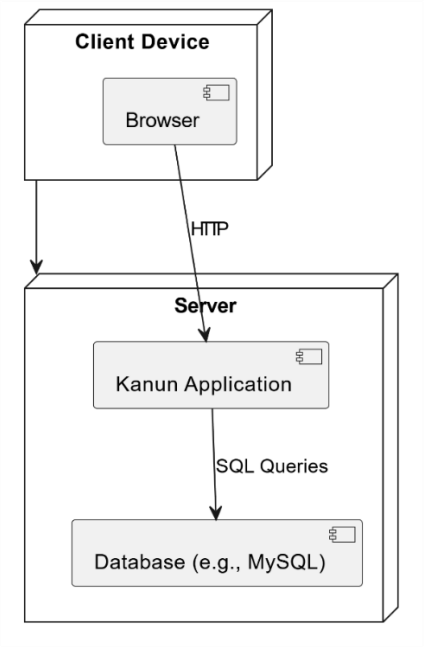


Figure 14: Deployment Diagram

The above diagram illustrates the deployment diagram for the Kanun application and relationship between the components. This system consists three main components include Client Device, Server and Database. Through client device such as computer or mobile which allows user to access Kanun application through browser through http protocol which send request to the server. The server hosts Kanun application for processes queries, retrieves relevant articles and generates responses from the constitution of Nepal. It uses the SQL queries to search for the important constitutional information. The database stores the constitution of Nepal to provide accurate responses while using the application

4.2 Algorithm Details

4.2.1 PDF to JSON Conversion Algorithm

PDF to JSON Conversion algorithm is used to convert the readable PDF file into JSON format

Algorithm:

Step 1: File Upload: First, allow users to upload PDF files to the Application

Step 2: PDF Processing: Extract the text from PDF(use pdfplumber)

Step 3: Match Pattern using Regex: The PDF file is matched with specific metadata such as(part, numbering, etc).

Step 4: JSON Conversion: Convert the matched pattern into JSON format

4.2.2 Algorithm for Chunking, Embedding and Injesting a Document into a vector Store

The algorithm requires JSON object that contains data about various parts, articles, clauses, explanations, and sub-clauses of a document.

Algorithm

Step 1: Chunk the Articles from the JSON:

Goal : Extract and organize the constitutional data into chunks of Articles.

Process:

- Initialize an empty list chunks.
- Loop through each part in the json_data:
- For each article in the "Articles" part:
- Start with the article's title as the base text.
- Loop through the clauses of the article:
    - Add the clause's text.
    - If an "Explanation" exists for the clause, append it.
    - If "Subclauses" exist, append their text as well.
- Append each chunk as a dictionary with a unique ID and text to the chunks list.

Step 2: Embed Text

Goal: Convert each chunk of articles into an embedding using a specific embedding model.

Process: Pass the text through the embedding model to generate an embedding(text-embedding-ada-002)

Step 3: Injest Document into Vector store

Goal: Convert the chunks into embeddings and store them in a vector store for future retrieval.

Process

- Store the chunk's id, embedding, and text in the vector store (or index).
- The vector store (or index) is updated using the upsert method.

4.2.3 Cosine Similarity Search Algorithm

Cosine similarity is a metric used to measure how similar the documents are irrespective of the size. Mathematically, it measures the cosine of the angle between two vectors projected in a multi-dimensional space. The cosine similarity is advantageous because even if the two similar documents are far apart by the Euclidean distance there will be a chance where they may still be oriented closer together.[34]

$$similarity(A, B) = cos(\theta) = \frac{A \cdot B}{||A||\,||B||} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2 \sum_{i=1}^{n} B_i^2}}.$$

The value of cosine similarity ranges from -1 to 1:

- 1 indicates that the vectors are perfectly aligned and point in the same direction (maximum similarity).
- 0 suggests the vectors are orthogonal (perpendicular), meaning they share no similarity.
- -1 shows that the vectors are diametrically opposed (point in opposite directions).

4.2.4 Retrieval Augmented Generation

RAG(Retrieval Augmented Generation) is a Generative AI architecture that enhances Large Language Models (LLMs) by combining information retrieval with generative capabilities, using fresh and trusted data from authoritative internal knowledge bases and enterprise systems to produce more accurate, contextually relevant, and up-to-date responses.[35]

Goal: To fetch the correct results from the vector store as per the user's query and Generate the answer using LLM capabilities.

Algorithm:

Step 1: Retrieve the query from the user.

Step 2: Preprocess the Query

- Trim the white space
- Convert the query into lowercase

Step 3: Validate the Query

- Check if the query contains at least two words.
- If invalid, return an error response.

Step 4: Embed the Query into Vector Form

- Use the `text-embedding-ada-002` model to generate vector embeddings.

Step 5: Retrieve Relevant Documents Using Vector Search

- Perform a cosine similarity search on the vector database.
- Retrieve the most relevant documents.
- Extract document ID and content from the retrieved documents.
- Format the Document ID and content into a structured context.

Step 6: Rephrase the Query (If Chat History Exists)

- Format previous chat history (if available).
- Insert the user query into a prompt containing chat history and context.

Step 7: Select Appropriate Prompt Based on Chat History

- If no chat history, use a prompt with an empty chat history.
- If chat history exists, use a prompt with chat history, input, and context.

Step 8: Generate Response Using LLM

- Feed the final prompt to the LLM (ChatOpenAI).
- Generate a response based on chat history, input, and context.

Step 9: Return the Response to the User

- Log the interaction (query, response, token usage, cost).
- Append the response to chat history and save the conversation.
- Render the response in the chat UI.

CHAPTER 5: IMPLEMENTATION AND TESTING

5.1 Implementation

A LangChain based application was developed to perform Retrieval-Augmented Generation (RAG) operations on the Constitution of Nepal. The development of the system was carried out in three phases. The first phase involved the integration of LangChain for building the retrieval system and incorporating preprocessors for handling constitutional data. The second phase focused on implementing the backend API and connecting it to the database for efficient data retrieval and storage. The final phase was the development of the frontend, which included the creation of a user-friendly interface for seamless interaction with the system that allows users to query the Constitution and receive accurate responses.

5.1.1 TOOLS USED

The tools used to create this we based visual demonstration of the proposed system are as follows:

5.1.1.1 Programming Language

i. HTML and CSS

HTML and CSS were used to design and styling the frontend for the project. HTML was very simple to edit as there was no need for a special interface or platform to edit. It can be simply edited in any editor and also has added flexibility and interactivity. It is used for developing and styling the structure of webpage, contents and layout.

ii. Python

Python played an important role in this project, particularly for implementation of LangChain Framework for performing Retrieval-Augmented Generation (RAG) operations on the Constitution of Nepal Python's extensive libraries and frameworks which enabled efficient development, testing and deploying such applications making it more powerful tools for both beginners and experts in the field of technology.

iii. Django Framework

Django was used for LangChain and RAG ensures that the application can effectively process and generate answers based on the Constitution of Nepal which made it a valuable tool for legal research and education. Django has the ability to handle database management, user authentication, API integration and security makes it an ideal choice for the project. User-friendly interface was created from Django which allows users to query the constitution and receive accurate, contextually relevant responses.

iv. SQLite3 Database

SQLite Database was used as the database to store constitutional data and user's data. It was chosen for its simplicity and ability to handle rational data efficiently in the embedded format. It provided a lightweight and reliable data storage solution, ensuring quick access to monitoring data while maintaining data integrity.

v. LangChain

LangChain was used in the system for performing Retrieval-Augmented Generation (RAG) operations on the Constitution of Nepal. It provides great utility for a variety of user cases from straightforward question-answering and text generation tasks to more complex solutions that use an LLM as a "reasoning engine".

vi. Visual Studio Code

Visual Studio code offers a comprehensive suite of tools and features that streamline the entire software development lifecycle, from coding and debugging to testing and deployment. It was used to write and manage the django framework and python scripts that handle RAG operations for the Constitution of Nepal.

vii. GitHub

GitHub was used to ensure proper changes and collaboration among the team members and to avoid code conflicts during the development processes.

viii. Draw.io

draw.io was used as diagramming tools for creating visual representations of the system components and relationships. Those tools were used for creating flowcharts, activity diagrams, class diagrams, and various other diagrams so that

they provide clear visualization of the various system components, workflow and interaction between the different components.

5.2 Testing

Software testing is a process of examining the functionality of a software which is under test to improve its performance and capabilities. This process helps in verifying and validating the actual software with the expected requirements to ensure that the product is built in the right way and is defect free. It involves identifying product bugs and requirements that have not been met. It also involves execution of software/system components using manual or automated tools to evaluate one or more properties of interest. The purpose of software testing is to identify errors, gaps or missing requirements in contrast to actual requirements.

5.2.1 Unit Testing

Unit testing is a software testing technique in which individual units or components of an application are tested in isolation from the rest of the application to ensure that they function as intended. In unit testing, each unit is tested separately to verify that it behaves as expected and produces the correct output for a given input.

5.2.1.1 Test Case 1

Table 9: Test Case For User Registration

Test Case ID	Test Case Description	Test Data	Expected Outcome	Pass/Fail Criteria
TC_REG_001	To verify the user gets registered with valid details.	Username:Ram Email: ram@gmail.com, Password: ram@1111	Users redirected to Login page	Pass
TC_REG_002	To verify the registrations whether it fails with multiple repetition of username.	Username:Raju Email: raju@gmail.com, Password:raju@123	Error message: A user with that username already exists.	Pass

TC_REG_003	To verify registration failure when the username input field is empty.	Username:``, Email: sita@gmail.com Password: sita@122	Error message: Please fill out field	Pass

Evidence for Test Case 1

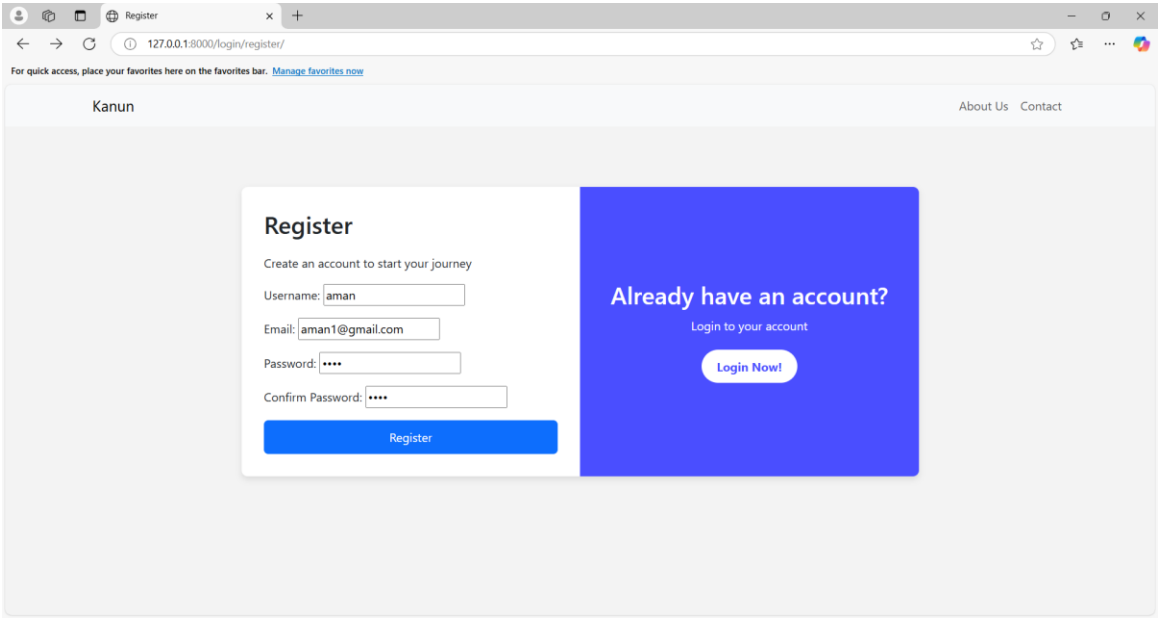


Figure 15: User creating a New Account

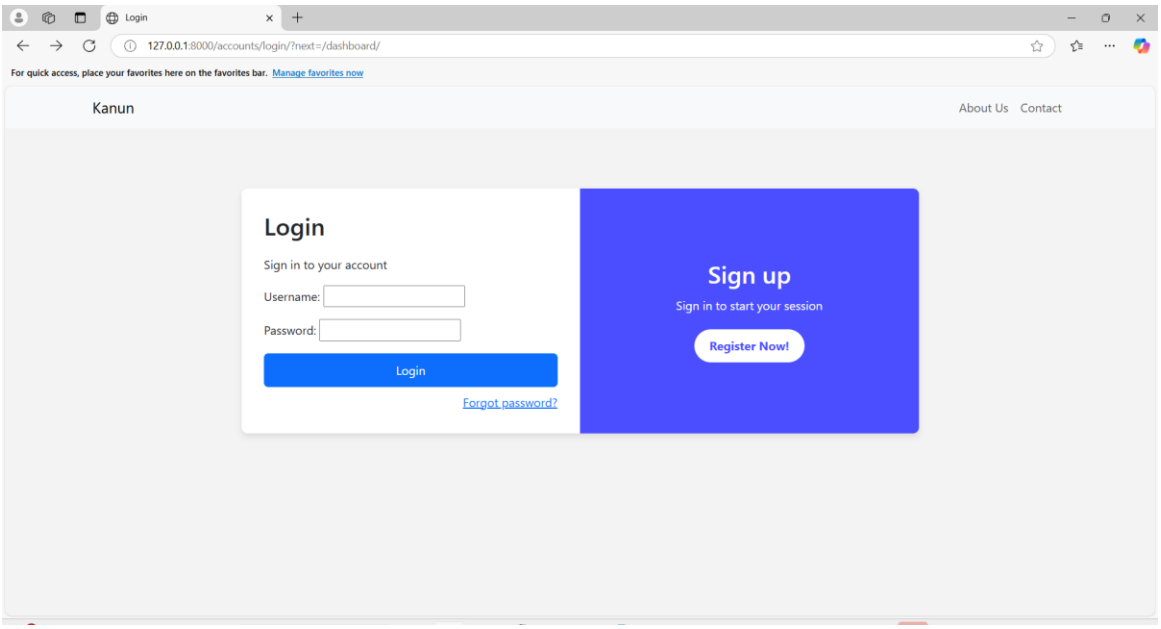


Figure 16: User redirected to Login Page

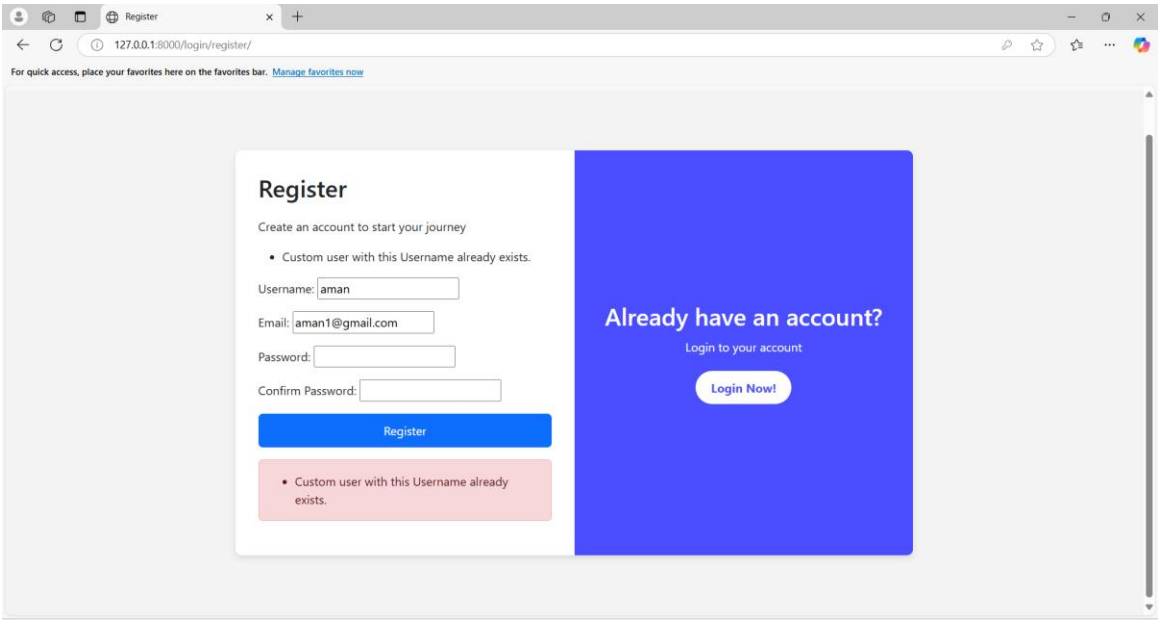


Figure 17: Error message popup with username already exist.

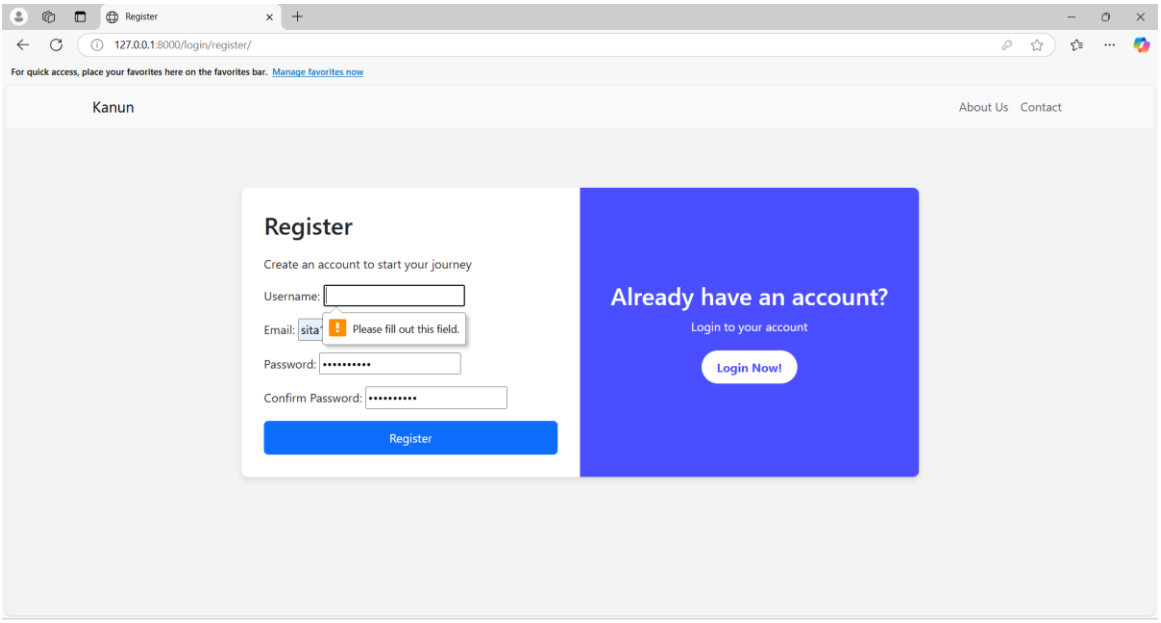


Figure 18: Error message popup with please fill out this field

5.2.1.2 Test Case 2

Table 10: Test Case For User Login

Test Case ID	Test Case Description	Test Data	Expected Outcome	Pass/Fail Criteria

TC_LOG_001	Verify that a user can Login with valid credentials.	Username: Ram Password: ram@1111	Users successfully log in. to dashboard	Pass
TC_LOG_002	Verify Login failure with an incorrect password	Username: Raju Password:raju@123	Please enter a correct username and password. Note that both fields may be case-sensitive.	Pass
TC_LOG_004	Verify Login failure when password field is empty.	Username: hari, Password:	Error message: Please fill out this field	Pass
TC_LOG_005	Verify Login failure whether both username and password field are empty.	Username: password:	Error Message: Please fill out this field	Pass

Evidence of Test Case 2

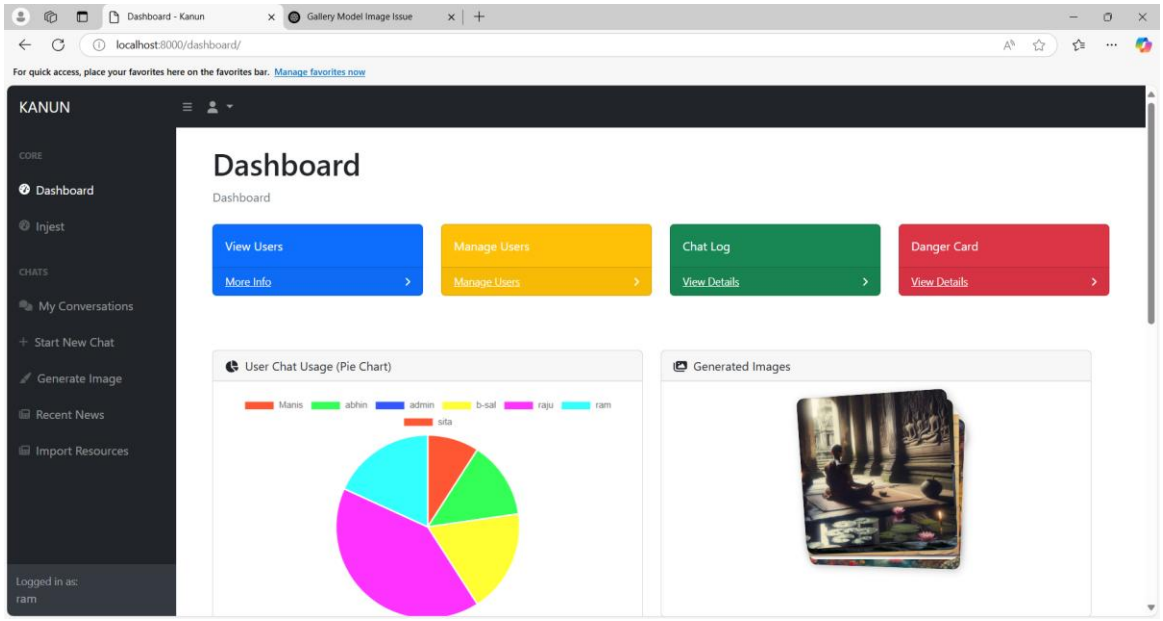


Figure 19: User successfully Login to dashboard

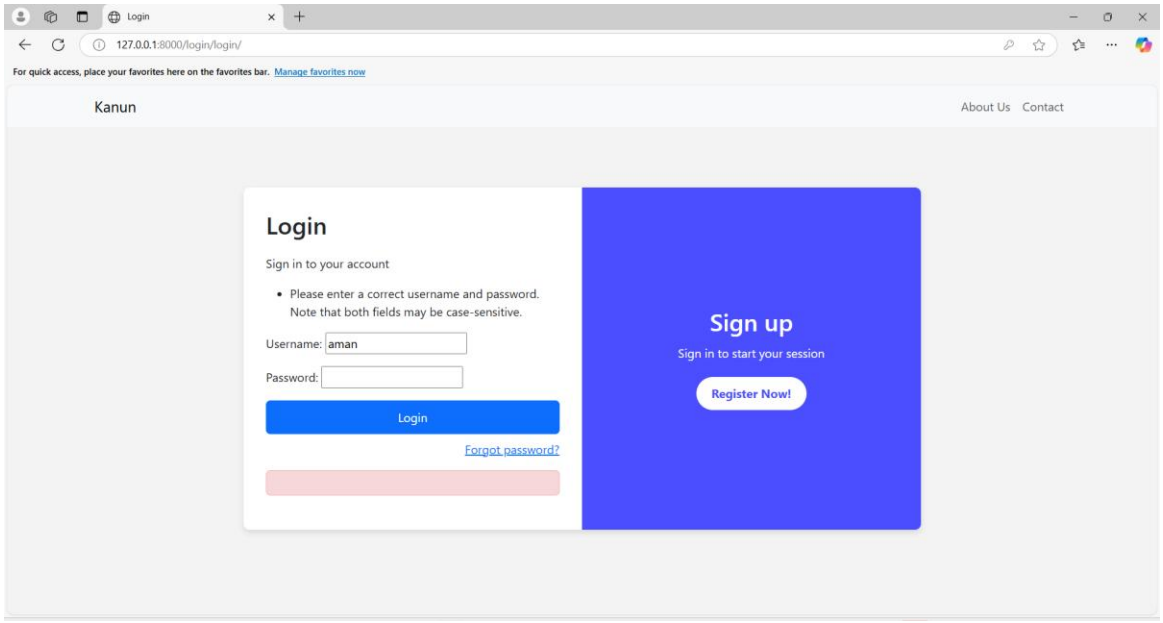


Figure 20: Login Fail due to invalid username and password

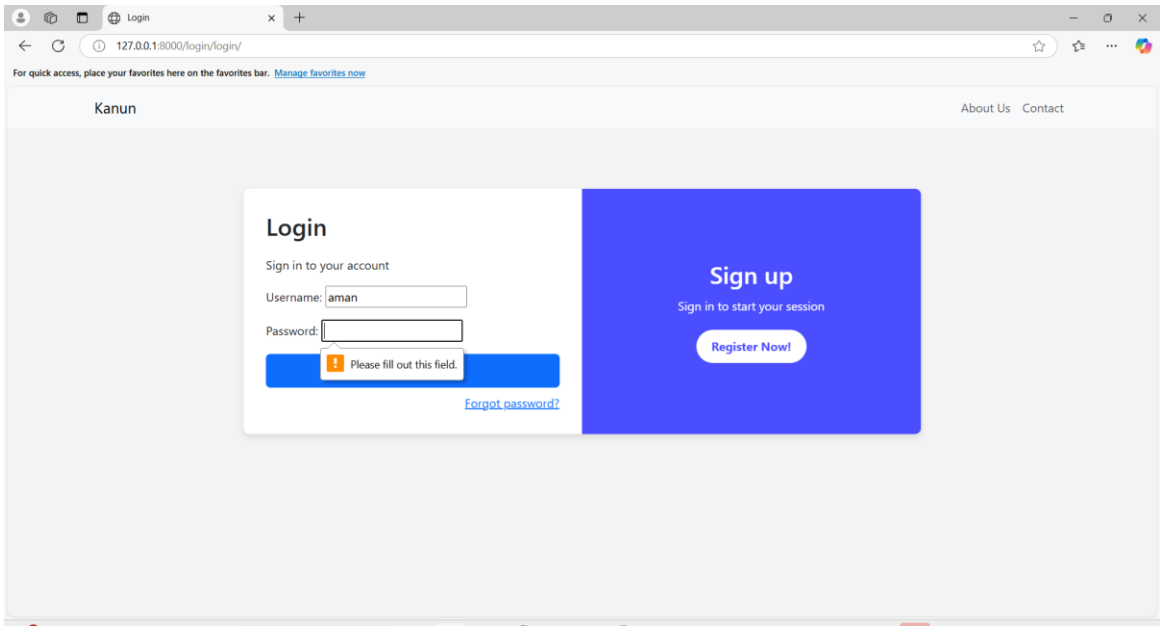


Figure 21:Error message popup with please fill this field

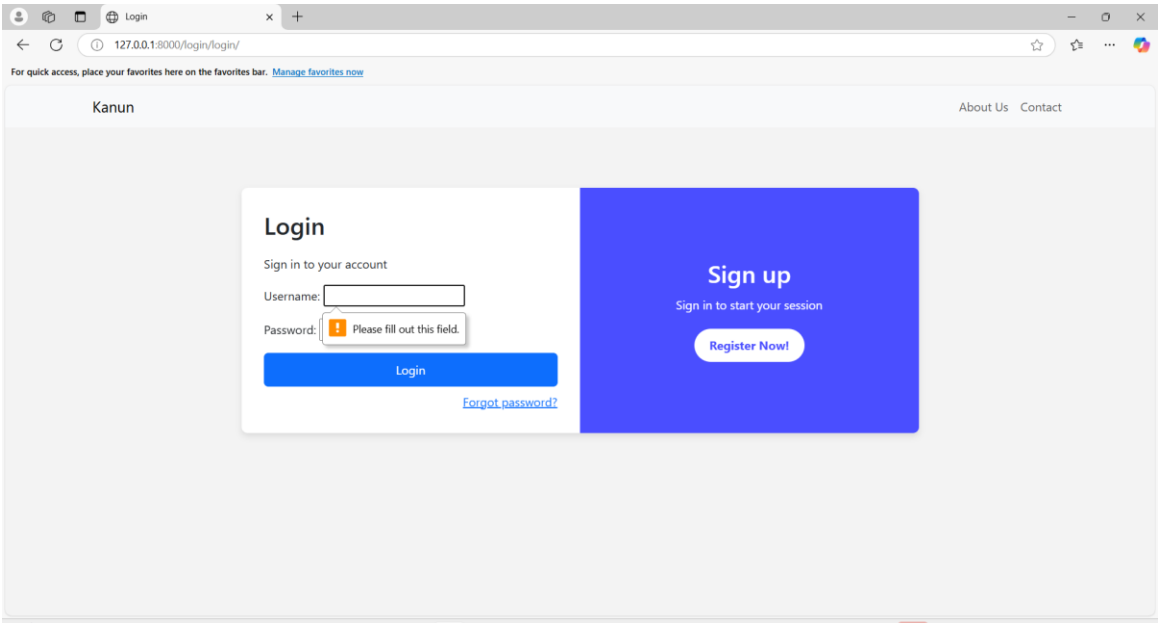


Figure 22: Error Message popup with please fill this field

5.2.1.3 Test Case 3

Table 11: Test Case For Chat Module

Test Case ID	Test Case Description	Test Data	Expected Outcome	Pass/Fail Criteria
TC_CHAT_01	Verify that a user can start a new chat.	Click on start new chat and type a question related to Constitution of Nepal	Chat initializes and the system responds with relevant constitutional information.	Pass
TC_CHAT_02	Verify that it answers the constitution related question or not.	Enter question about Constitution of Nepal	System should reject the query with an “ I don't know how to respond ” message.	pass.
TC_CHAT_03	Verify chat history was saved.	Start the conversation and ask multiple questions.	The chat history should display previous queries and responses.	Pass

TC_CHAT_04	Verify the real-time response generation.	Ask the constitution related question	System should generate and response within a reasonable time	Pass
TC_CHAT05	Verify responsiveness for mobile and desktop	Access chat modules from different devices.	Chat interface should adjust for screen size	Pass
TC_CHAT_06	Verify logout clears chat history and saved into my conversation	Logout and login again	Previous chat history should be visible in my conversation.	Pass

Evidence of Test case 3

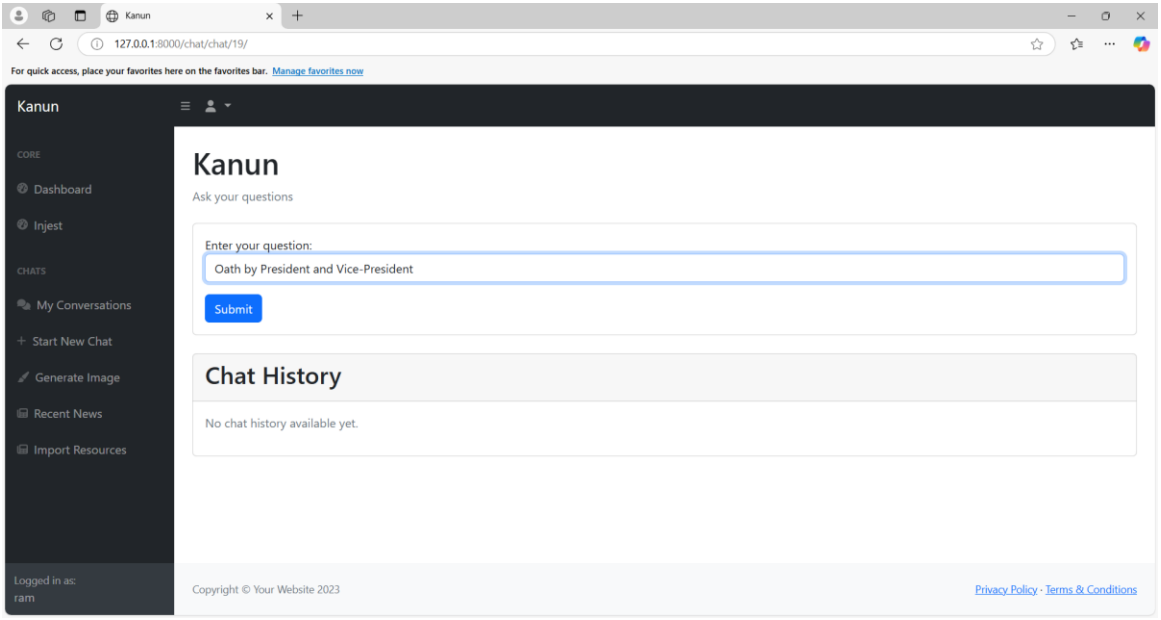


Figure 23: Giving Query to the KANUN

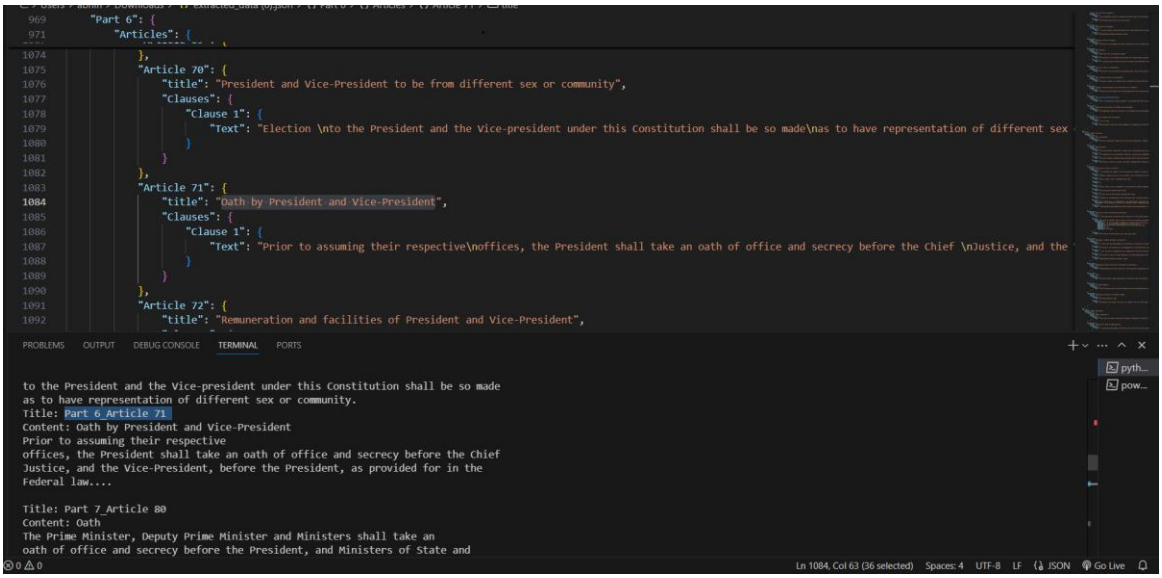


Figure 24: Fetched Content From vector database

Figure 24: Fetched Content From vector database

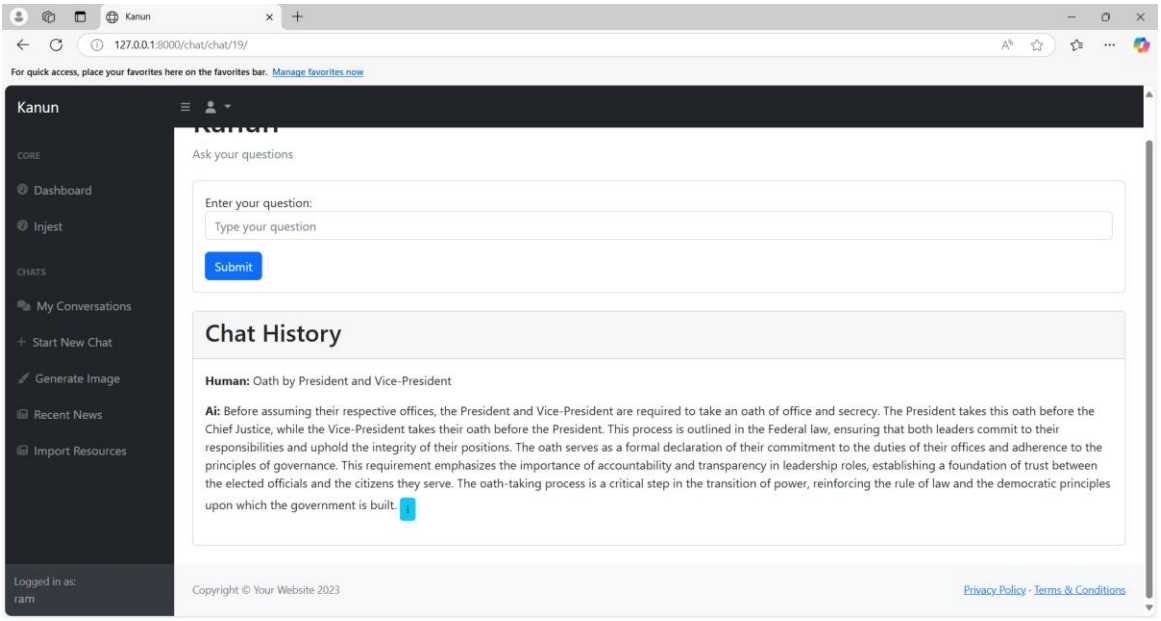


Figure 25: Generated Response

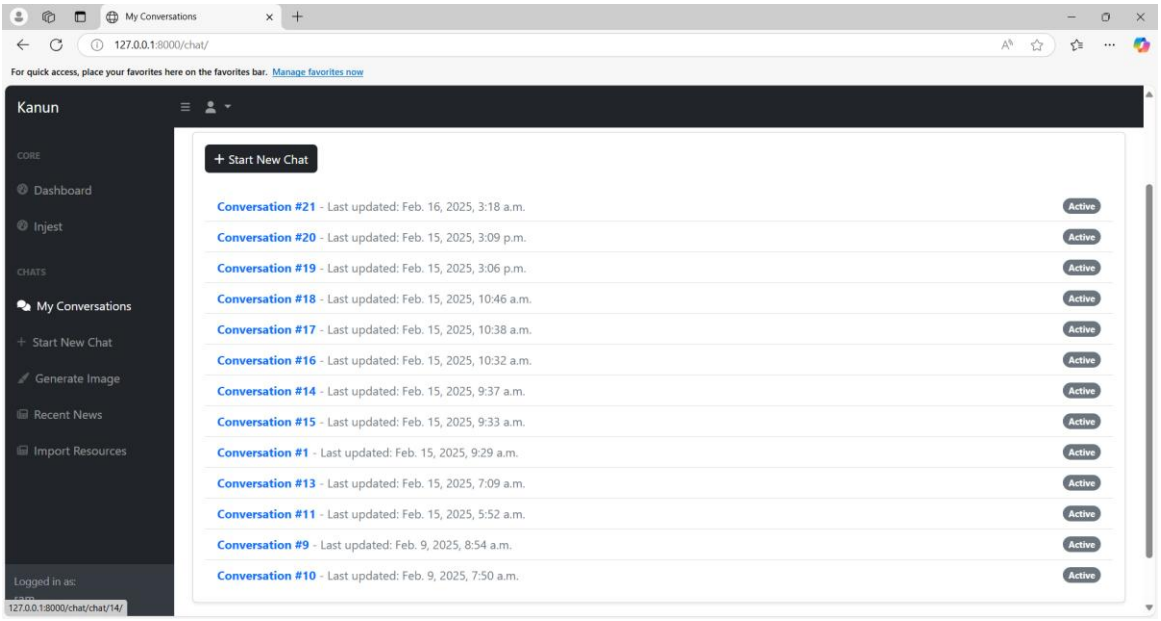


Figure 26: New Chat was created

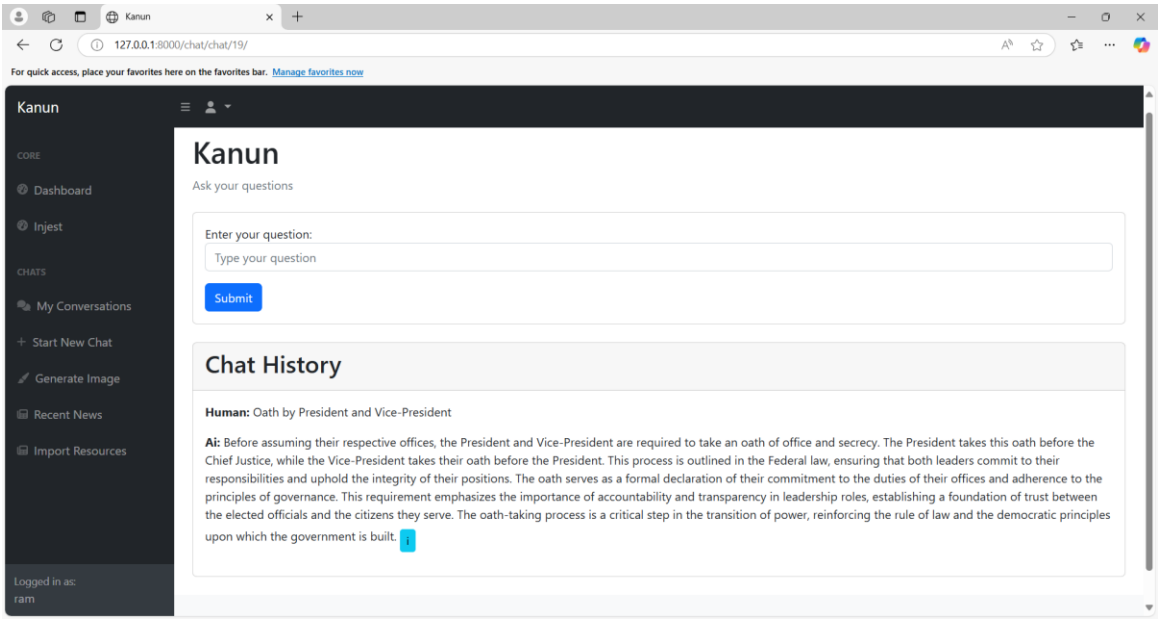


Figure 27: The conversation was saved in chat History

5.2.2 System Testing

System testing is a software testing technique used to evaluate the behavior of an entire software system or application, typically by testing the system as a whole rather than individual components or modules. The goal of system testing is to ensure that the software system works as intended, meets the specified requirements, and is free from defects and bugs that could negatively affect its functionality or usability

Table 12: System Testing

Test Case	Test Scenario	Steps to Execute	Expected Result	Actual Result	Status
Login	Verify that user can login successfully	1.Go to the Login 2. Fill out the form with valid details 3.Display dashboard to user	The user should be logged in and redirected to their dashboard.	Successfully login to the user dashboard	Pass
Registration	Verify that user can	1.Go to the /sign up or	The users should be registered and	Users successfully	Pass

	register successfully	/register page. 2.Fill out the form with valid details. 3.Submit the form	see a success message.	register the application.	
User Management	Verify that staff can access the details	1.Login with staff valid details. 2.Go to the user management 3.Details of valid users.	The staff user should be accessing the user details with the edit, delete and add button.	Staff users successfully get access to user details .	Pass
Chat Module	Verify that user can start new chat	1.Click on the start new chat 2. Type related queries to Constitution of Nepal	The user can initialize the chat and respond with relevant constitutional information.	User successfully get respond with relevant constitutional information	Pass

5.3 Result Analysis

Result analysis is the process involved in software development which involves the evaluation of the outcomes of various stages in the software development life cycle to ensure that project meets the objective and targeted requirements.

5.3.1 Proposed Model

The proposed model for Kanun application is a secured and legal information retrieval system that works on Retrieval-Augmented Generation (RAG) concept. The application is designed to help the user to retrieve the most relevant and accurate legal information. The Constitution of Nepal has been preprocessed into chunks which are converted into vector embedding using a pre-trained language model of OpenAI. A generative model then synthesizes these retrieved sections into coherent, human-readable responses and contextual explanation.

The Django framework as the backend handles the API requests and responses where a responsive frontend allows users to interact with the system via a web browser. This application enhances accessibility to legal information which makes it easier for citizens to

get through the Constitution of Nepal including those without legal expertise to understand human rights and obligations. In Nepal, the proposed of Kanun application could help to acknowledge the citizen right toward the Nation.

Additionally, the use of RAG technology could increase the transparency and people can access rules and regulations as they need. The Application provides a powerful tool for citizens, legal professionals and policymakers in Nepal. However, implementing Kanun application using RAG technology in Nepal would require overcoming several challenges, such as ensuring secure and reliable internet connectivity and building trust among citizens in the new application.Therefore, careful planning and collaboration among stakeholders, including government agencies, civil society would be essential to successfully implementing the proposed Kanun application in Nepal. Kanun had the potential to revolutionize how people interact with the Constitution of Nepal allowing them to recognize their rights and fulfill their responsibilities with full confidence against any crime.

5.3.2 Advantage of Proposed Model

The proposed Kanun query application using RAG technology offers several advantages over the traditional learning systems. It ensures that responses are precise and accurate in the original text. A user-friendly interface designed by Django makes it easier for the people and the citizens of Nepal to submit queries and receive detailed information about the Constitution of Nepal. Citizens can learn their legal rights and values from anywhere in the world, making the process more accessible and convenient. The application can reduce the costs associated with the traditional learning systems. The need for books, legal advisors and libraries is eliminated, reducing the overall cost of the traditional learning system.

5.3.3 Disadvantages of Proposed Model

As the application is based on RAG, it requires a large database of pre-processed data, which can be an extensive resource commitment. Similarly, in order to store the embedded data of the documents requires complex setup of vector database and interactions are complex that can cause additional latency in the system. It requires specialized expertise for development and maintenance. Furthermore, the accuracy of responses hinges on the quality and completeness of the underlying data. If the training dataset is flawed or incomplete, the AI may generate inaccurate or misleading information, potentially leading to user confusion and misinterpretations of constitutional articles. This reliance on data

quality also highlights the critical importance of robust data governance and curation processes. Finally, the handling of sensitive legal queries raises serious concerns about data privacy and security, demanding stringent safeguards to protect confidential information and prevent misuse. Addressing these challenges is crucial for the responsible and effective deployment of AI-powered legal tools.

5.3.4 Relevant Screenshot

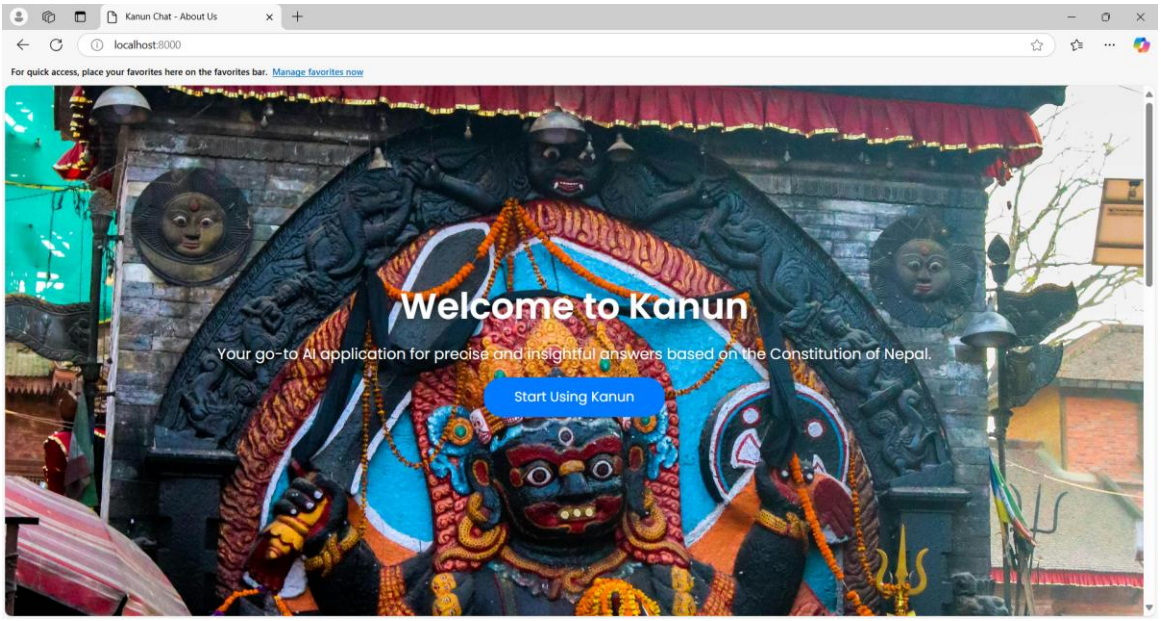


Figure 28: Cover Page

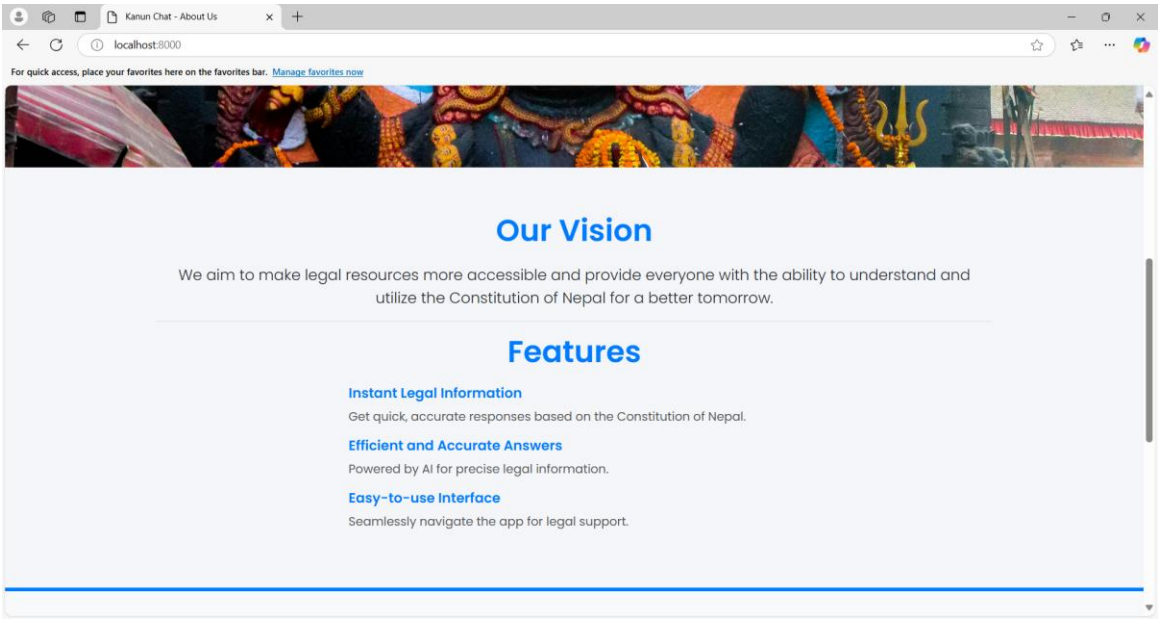


Figure 29: Cover Page - Introduction

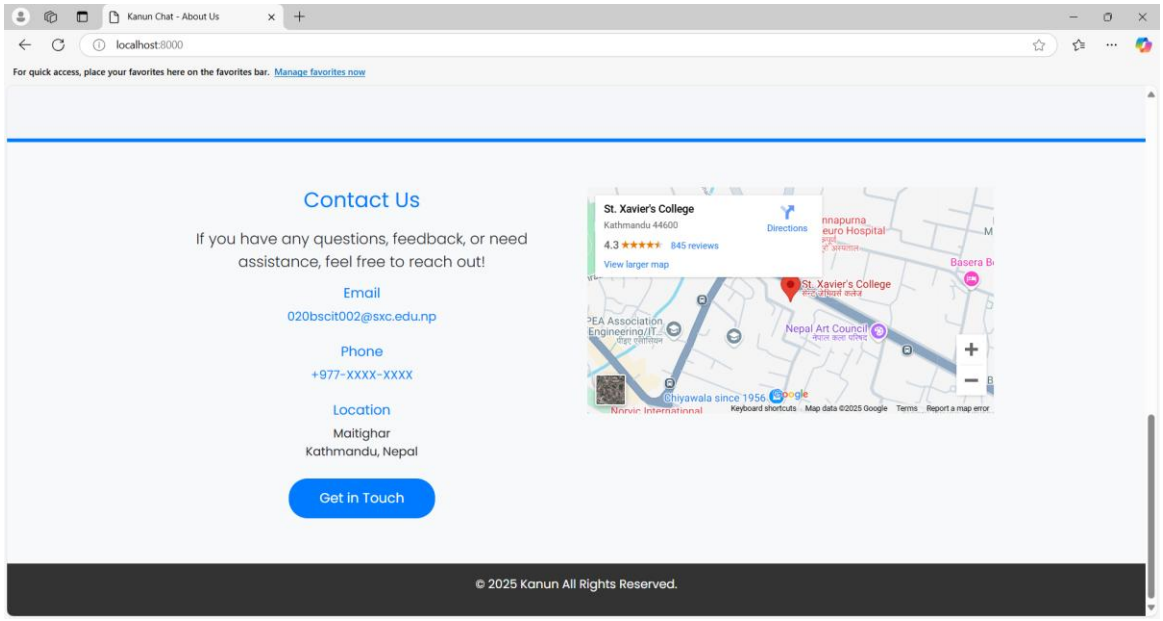


Figure 30: Cover Page - Contact

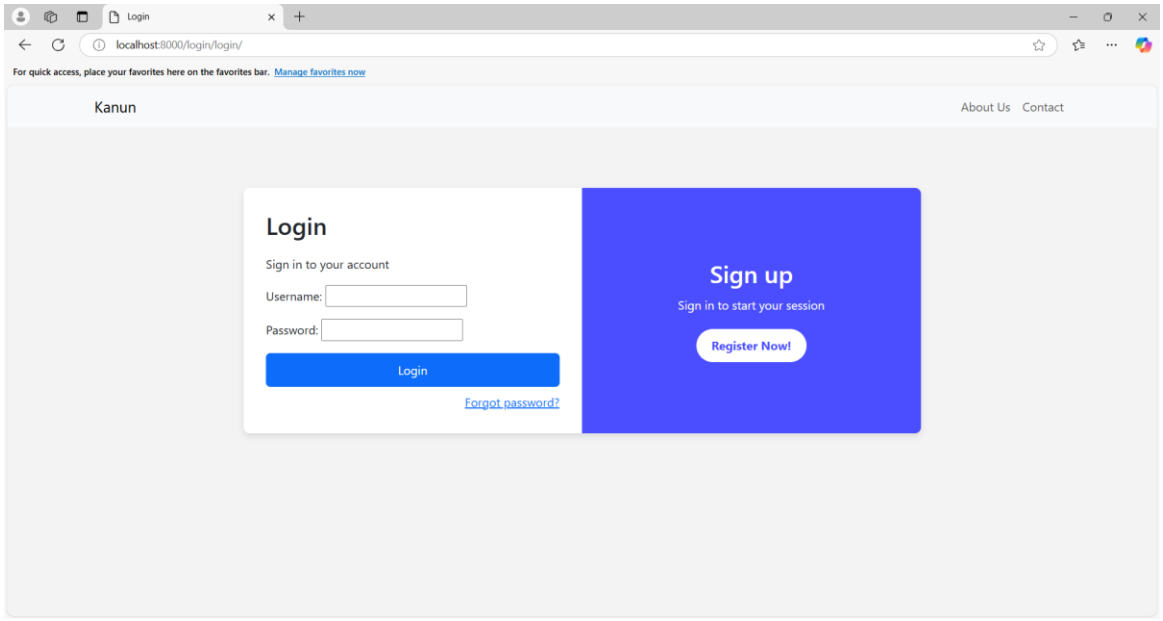


Figure 31: Login Page

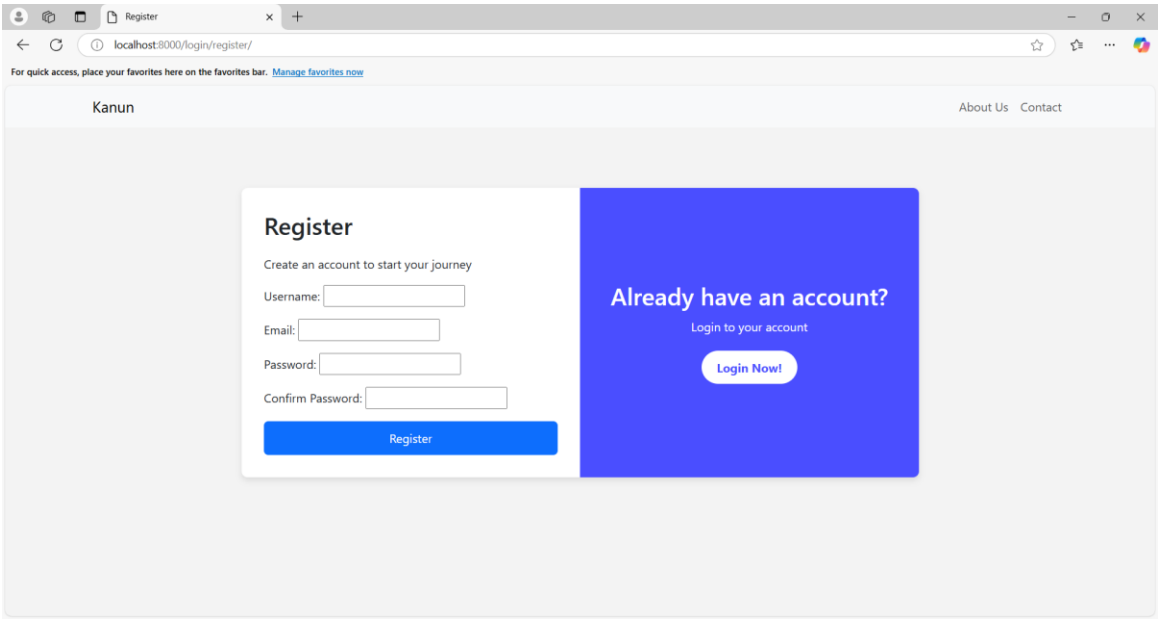


Figure 32Registration Page

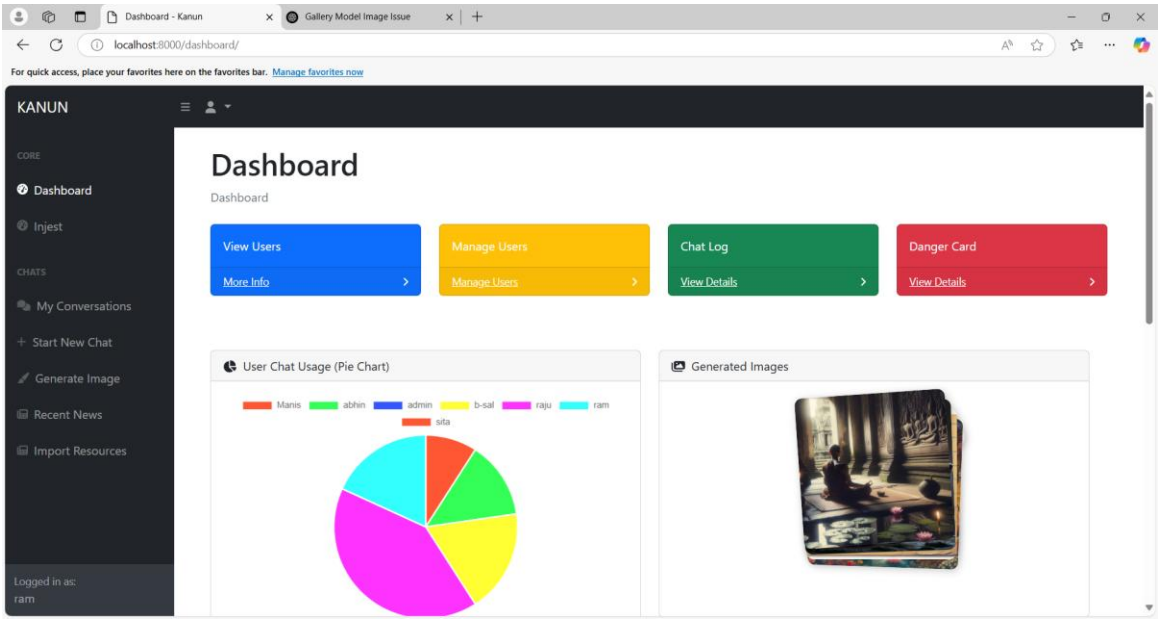


Figure 33: Dashboard Page

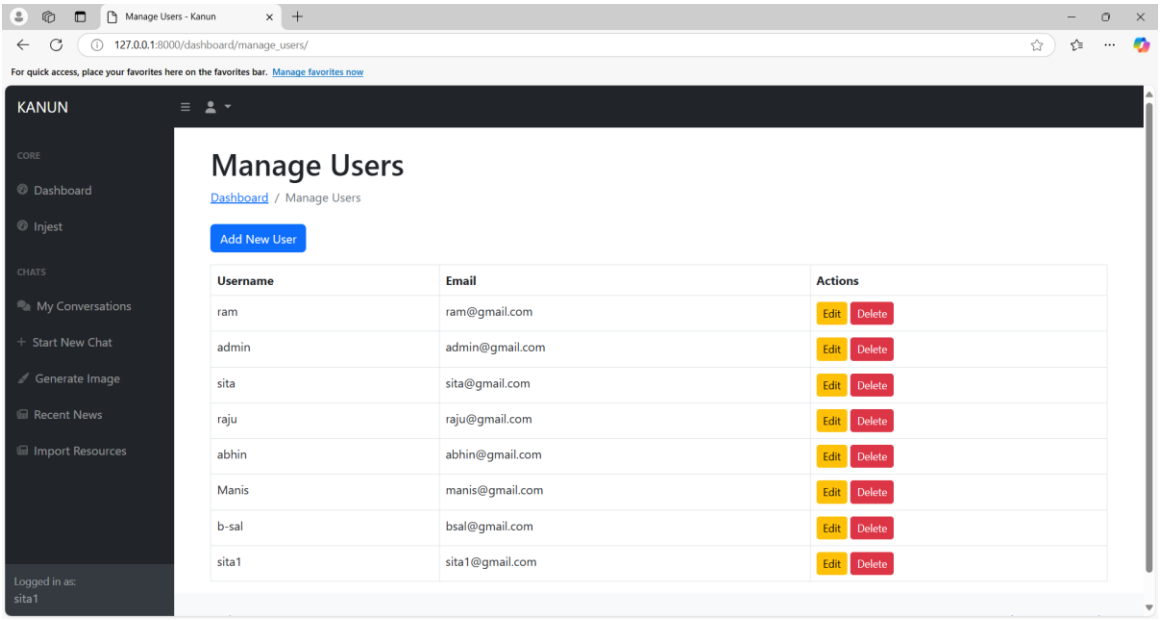


Figure 34: User Management

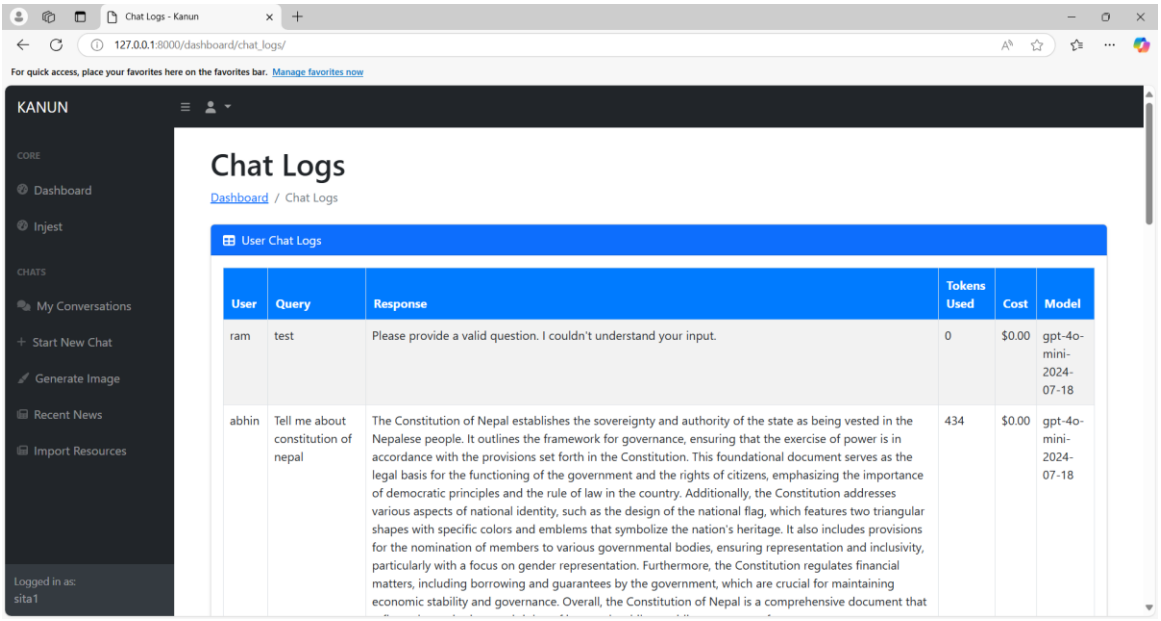


Figure 35: Chat Log

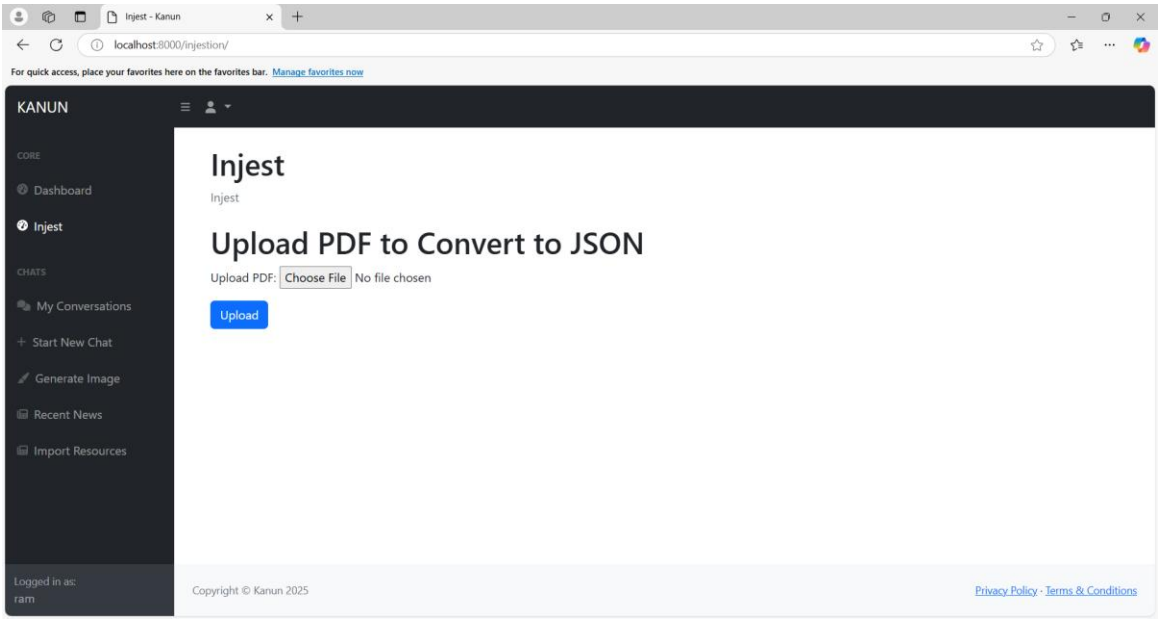


Figure 36: Injestion Page

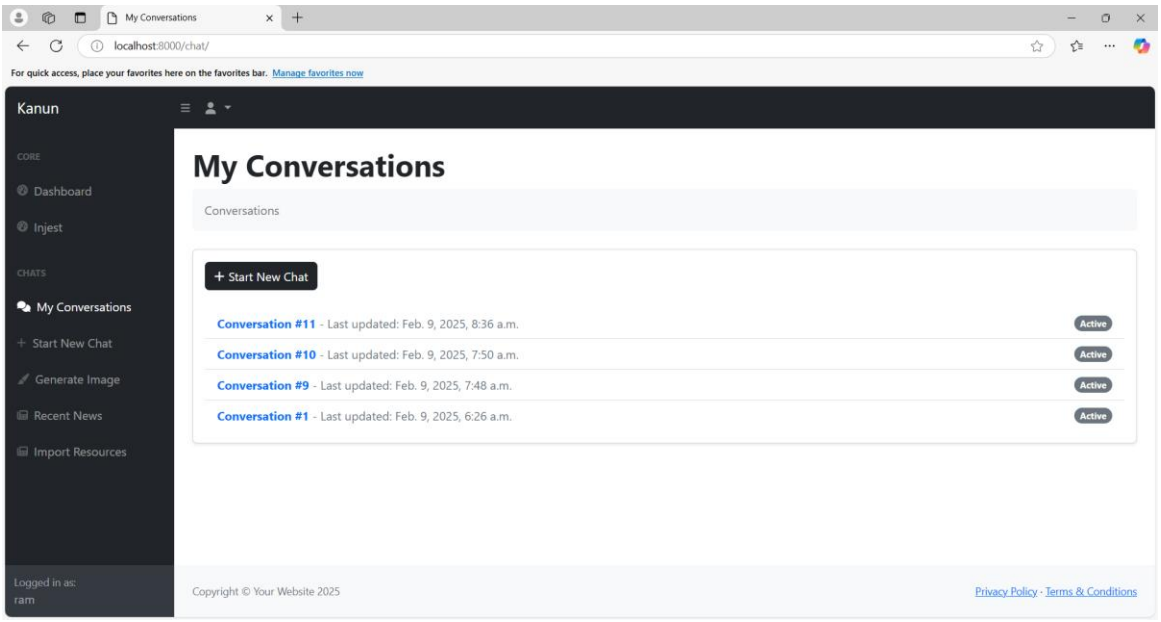


Figure 37: Conversation Page

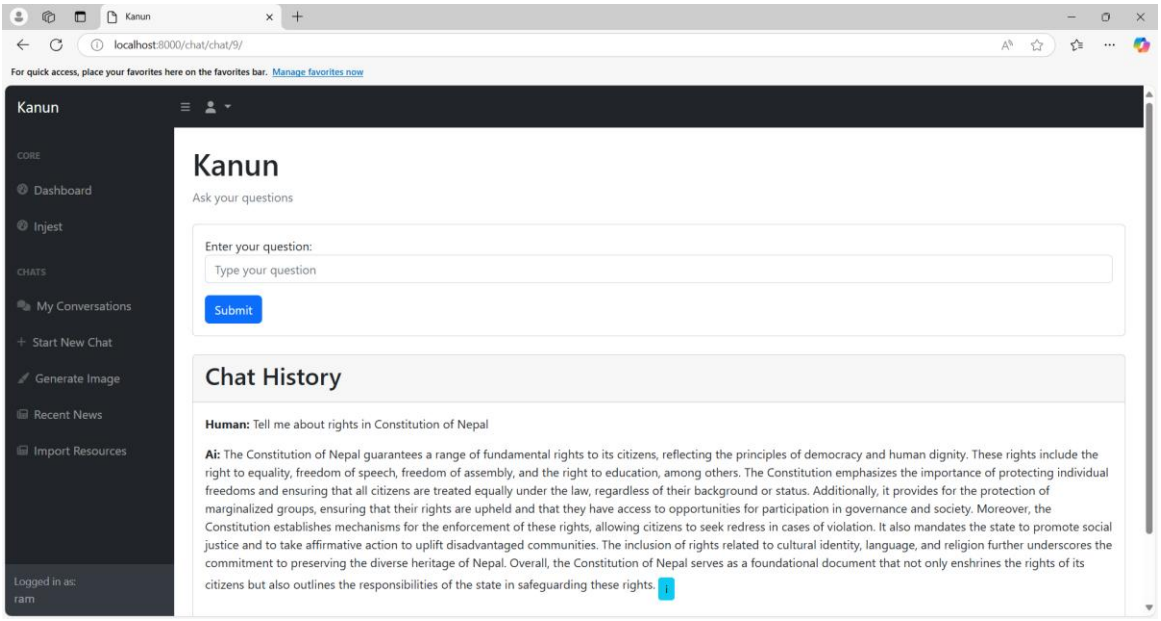


Figure 38: Chat PAge

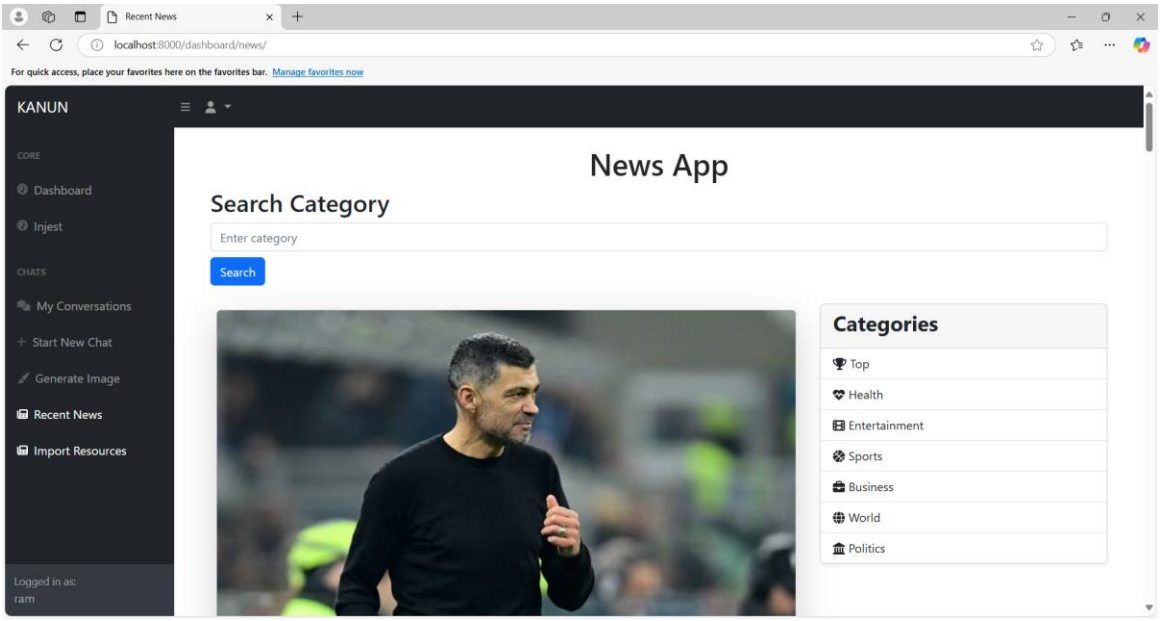


Figure 39: News Page

CHAPTER 6: CONCLUSION AND FUTURE RECOMMENDATION

6.1 Conclusion

The "कानुन" Kanun application, powered by Retrieval-Augmented Generation (RAG) technology for Nepal's Constitution. It is designed centered around pre-processed constitutional text, whichis feed to LLM through RAG and allows for the generation of clear and contextual responses to user inquiries. This approach aims to bridge the gap between complex legal language and the everyday citizen, fostering a deeper understanding of fundamental rights and obligations. The Django-based interface further simplifies interaction also this enhanced accessibility has the potential to cultivate greater transparency and citizen's participation and empowering individuals to navigate the legal framework with increased confidence. While the benefits are clear, including improved accuracy and cost-effectiveness compared to traditional legal resources, certain implementation hurdles must be addressed. These include the considerable effort required for data preparation and storage, the technical expertise needed for ongoing maintenance, and the paramount importance of data integrity for ensuring reliable outputs. Similarly, data management and security protocols are vital to safeguard sensitive information and build public trust. Overcoming these challenges is essential to fully realize the transformative potential of the Kanun application and its impact on legal literacy in Nepal. Despite these complexities, the Kanun application offers a promising avenue for reshaping the way citizens engage with the Constitution, ultimately fostering a more informed and empowered populace.

6.2 Future Recommendation

Looking forward, several important recommendations are essential for the ongoing growth and enhancement of the Kanun application. Considering multilingual support is necessary to increase accessibility for a broader audience. Furthermore, incorporating advanced retrieval methodologies, such as Recursive Abstractive Processing for Tree Organized Retrieval, has significant promise for enhancing the system's capability to efficiently explore and summarize intricate legal information. Developing a mobile application, along with enabling offline access to vital information, is crucial for connecting with users in remote locations that have limited internet access. Implementing a strong user feedback system will be invaluable for collecting insights and steering future development initiatives.

Lastly, promoting collaboration with legal experts, government entities, and local communities is key to ensuring the accuracy, relevance, and overall effectiveness of the Kanun application.

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